

Tutorial

Artificial Intelligence in

Microfluidic Circuits

Speakers

Jorge Torres Gomez, Murat Kuşçu, Cansu Canbek, Volkan Can

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Memorial University

Canada

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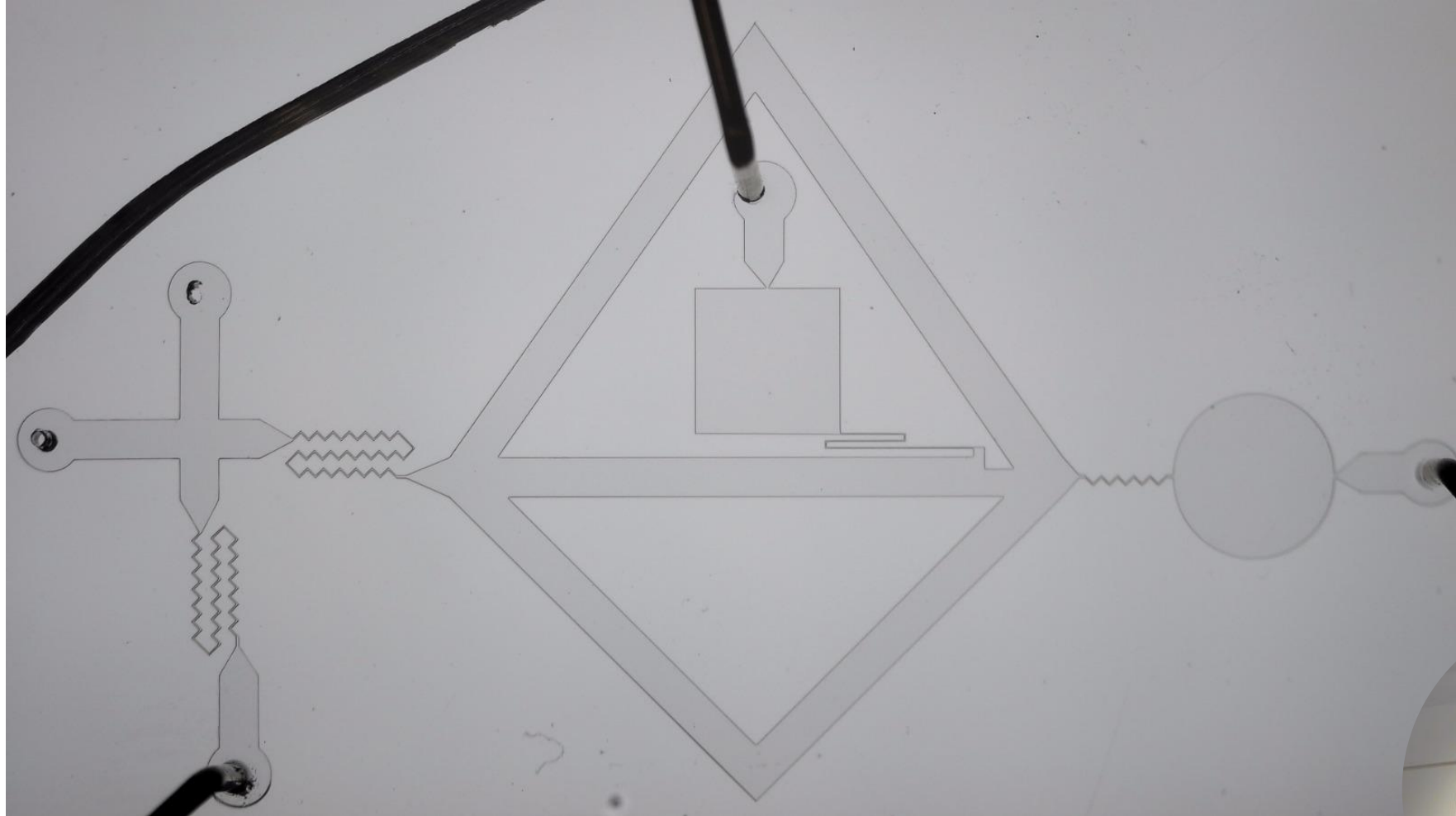
DSP in Microfluidic MC Systems

Summary

- Introduction
- Analysis of microfluidic flows
- Synthesis of microfluidic designs in COMSOL



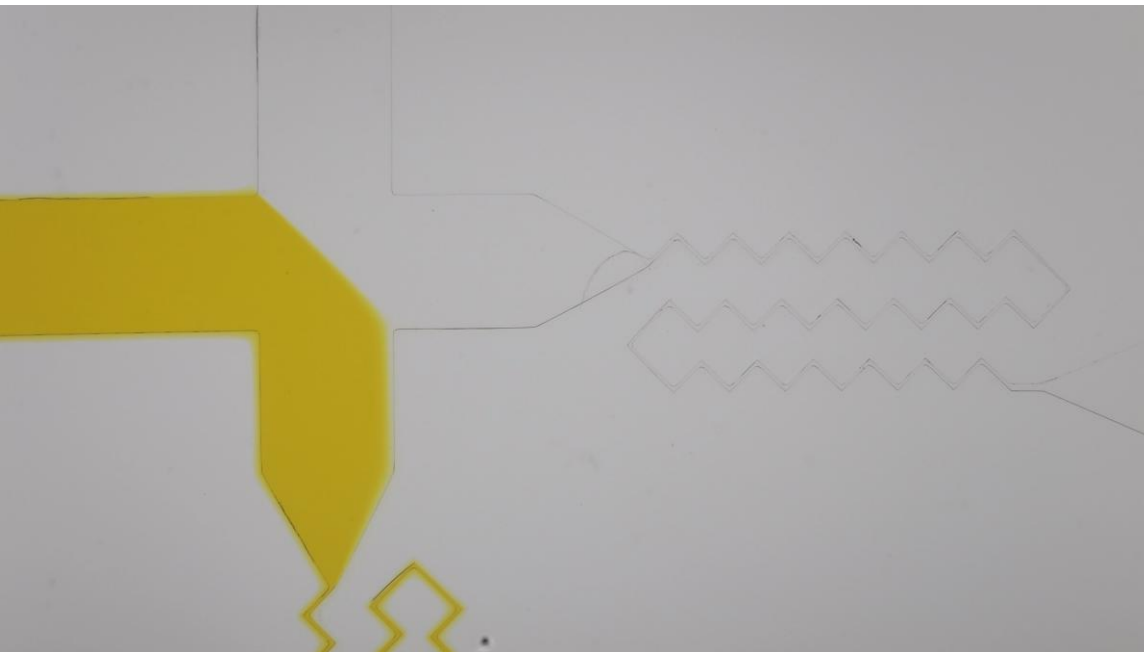
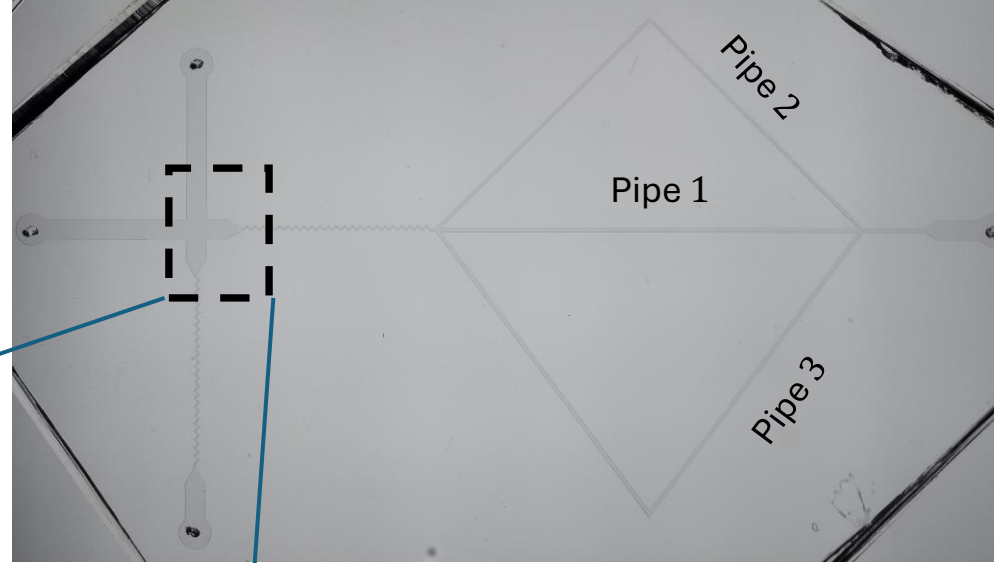
Microfluidic Circuit

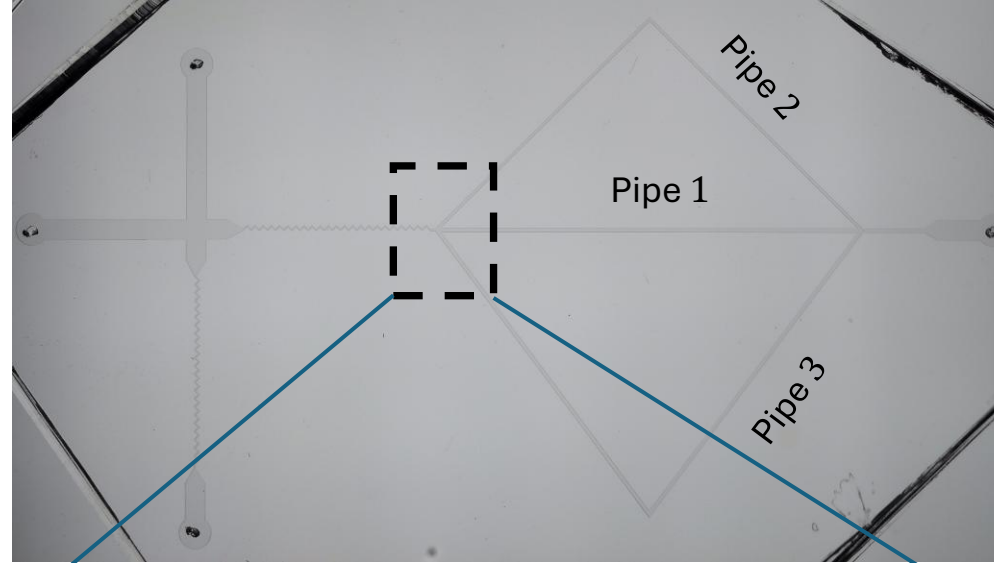


19 cm

35 cm

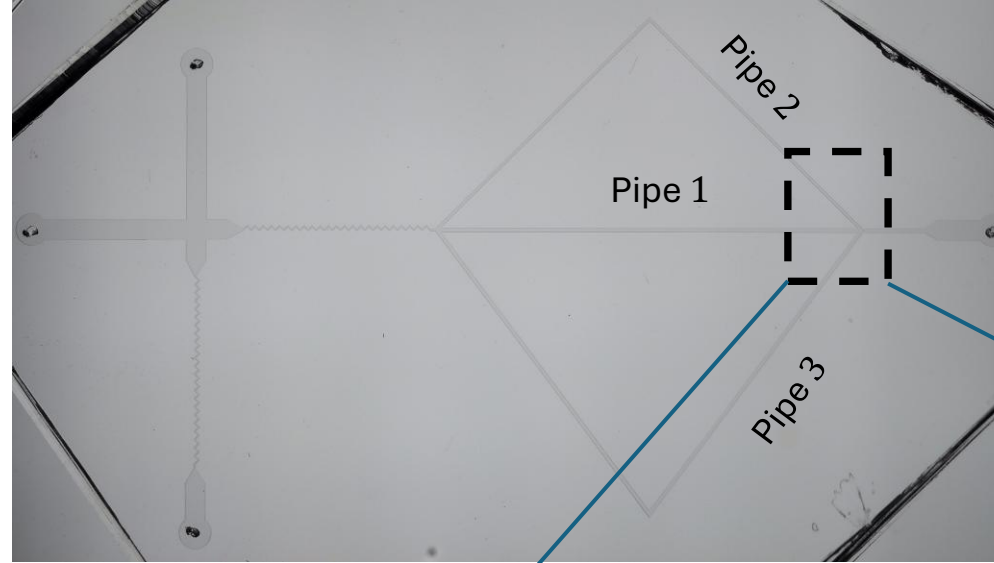






Pipe 2





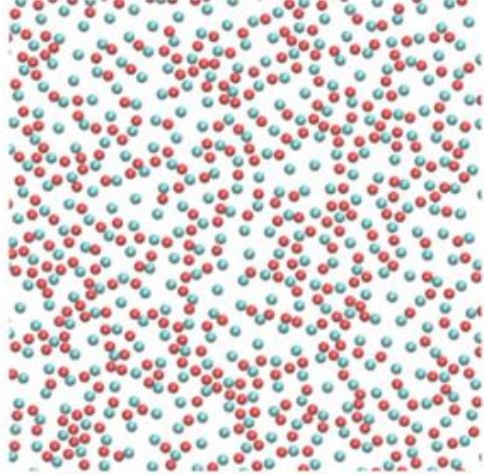
Pipe 2



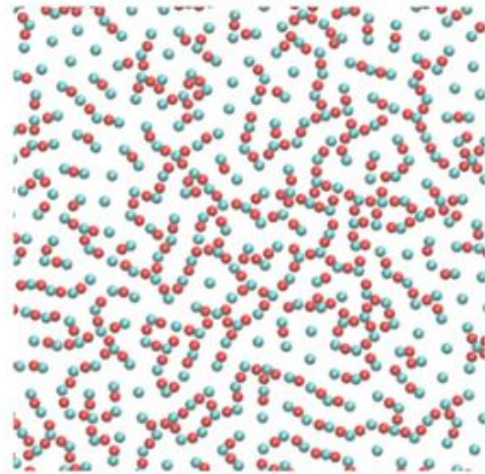
Volumetric Flow: $Q < 20 \mu\text{L}/\text{min}$

Traveling time: seconds – minutes

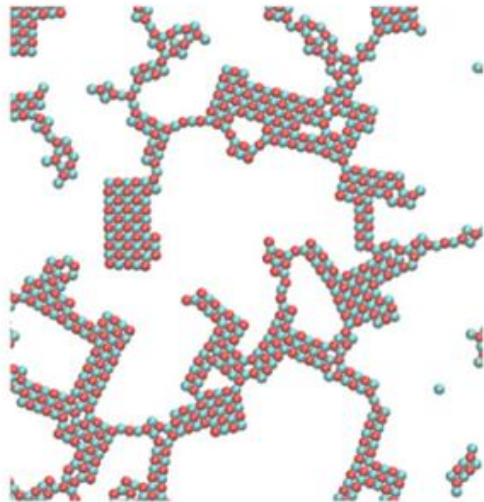
A Static potential (pH = 3)



B Static potential (pH = 4)



C Static potential (pH = 5)



D Dynamic potential
pH 3 → pH 7 → pH 3



Chemical pH Oscillators [1]

Self-Assembly

Periodic drug delivery

Molecular Motors

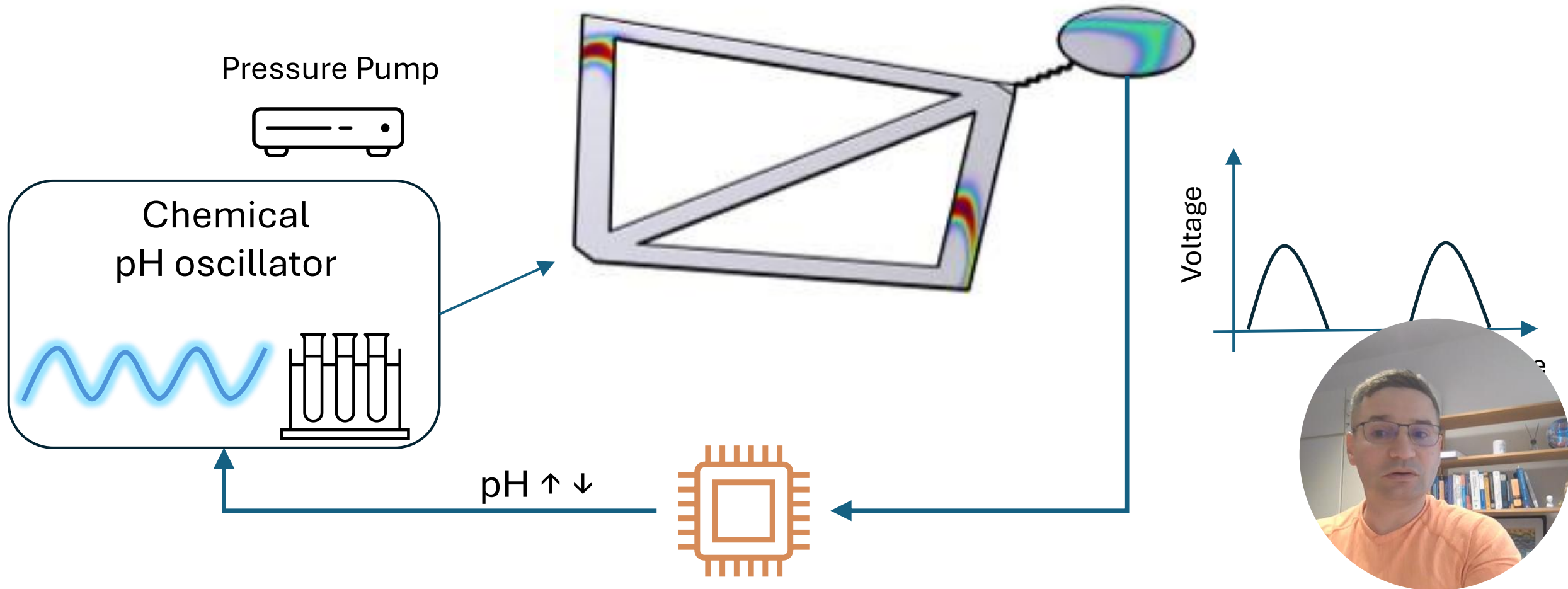
DNA-based nanodevices

Fast-tracking Applications

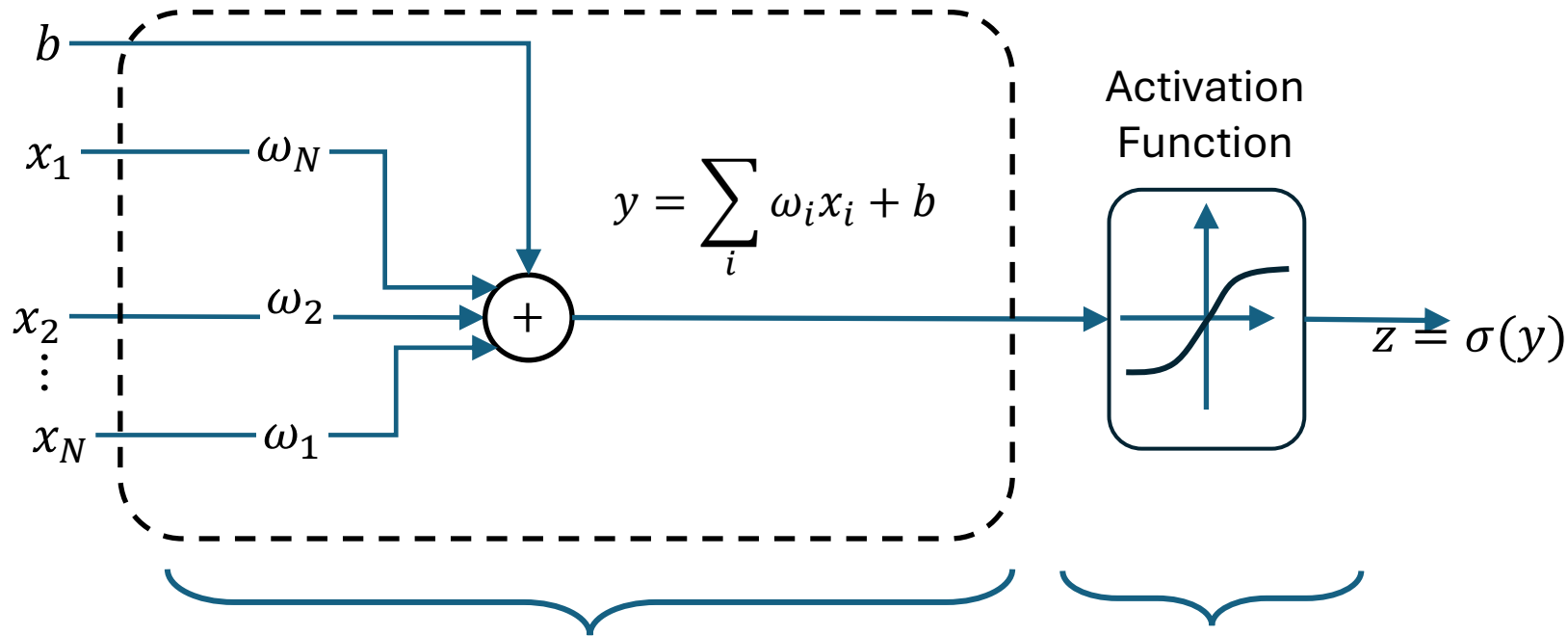
- M. Orbán, K. Kurin-Csörgei, and I. R. Epstein, "pH-Induced Self-Assembly in Chemical Oscillators," *Acc. Chem. Res.*, vol. 48, no. 3, pp. 311–318, Mar. 2015, doi: [10.1021/ar5004237](https://doi.org/10.1021/ar5004237).



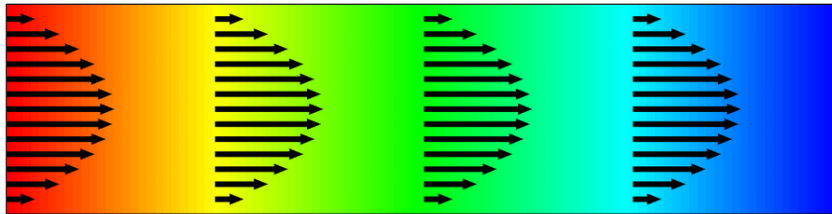
Application context



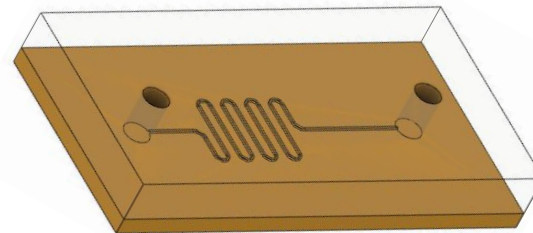
Single Neuron Scheme



Transport of Diluted Species



Chemical Reaction

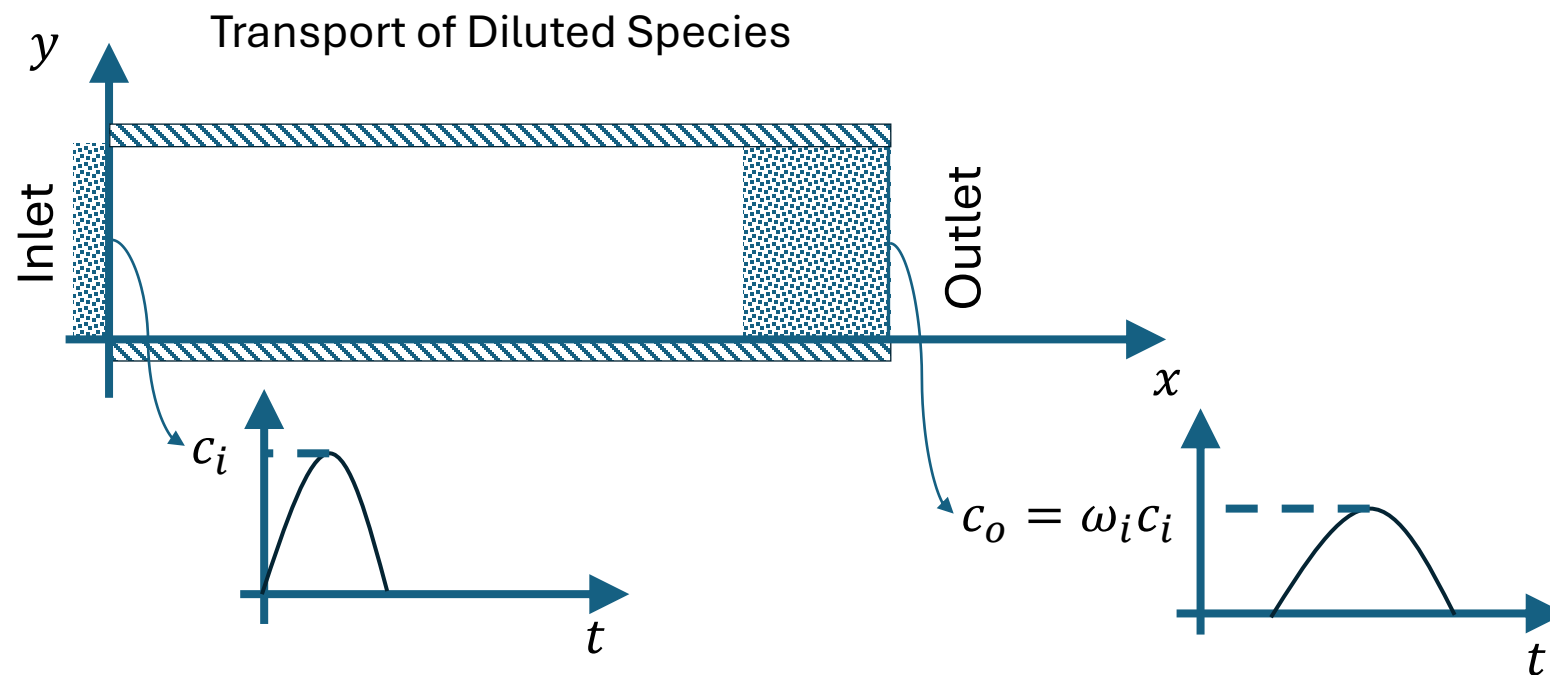
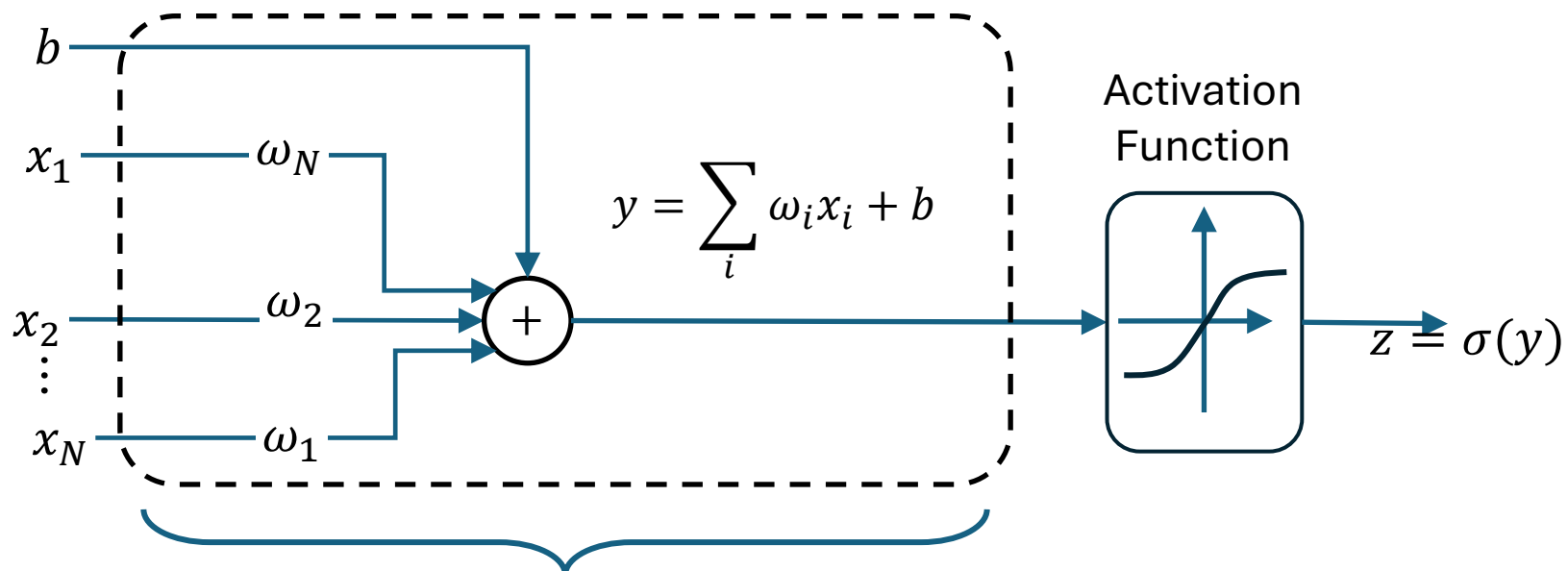


Microfluidic Circuit

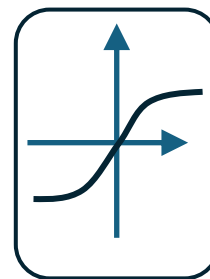


[1] A. O. Bicen
Theoretic Analysis of
Microfluidic Channels for
Molecular Communication
on Signal Processing
5013, Oct. 2013, doi:
10.1109/TSP.2013.2274959.

Single Neuron Scheme

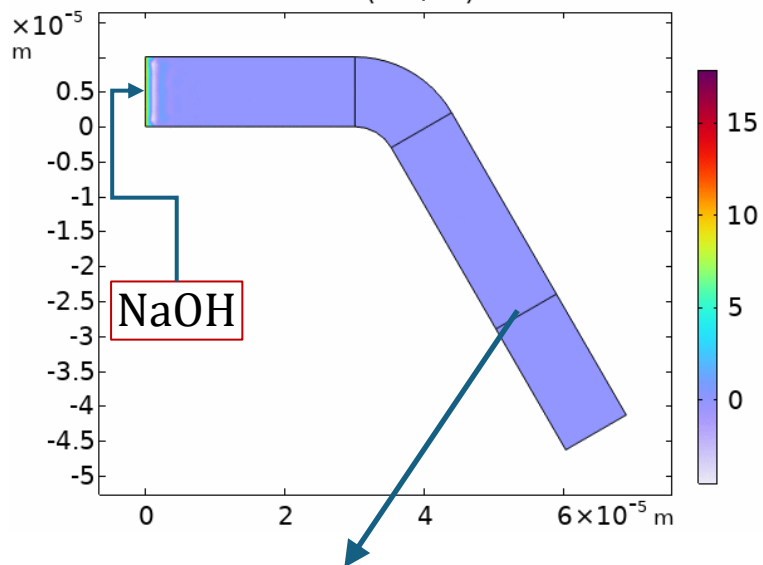


Concept

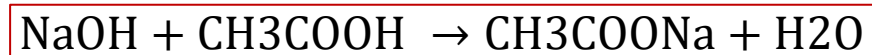
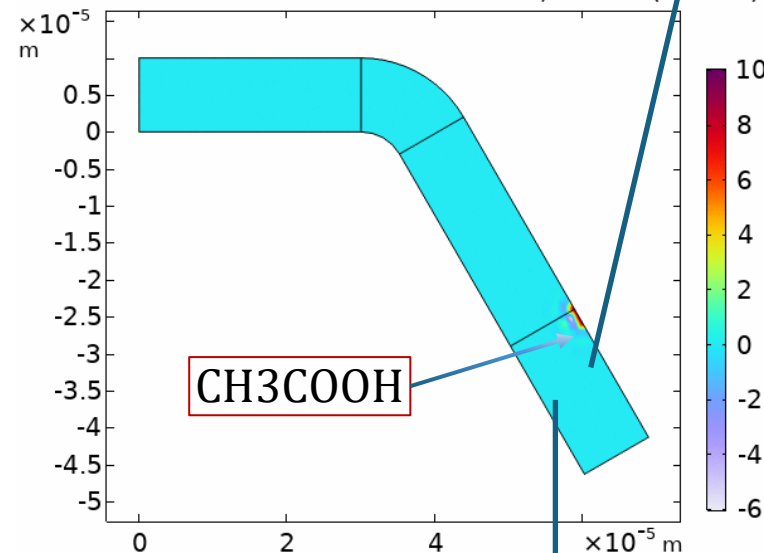


Activation
Function

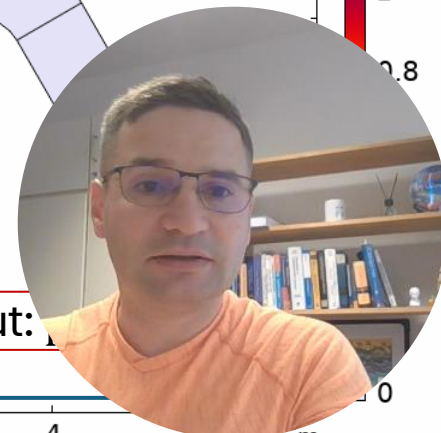
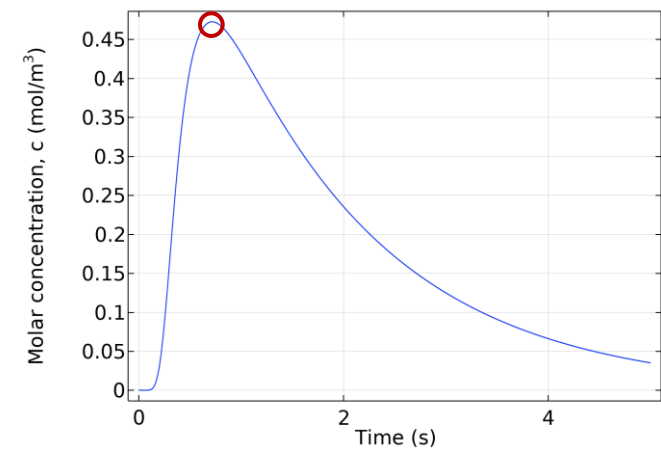
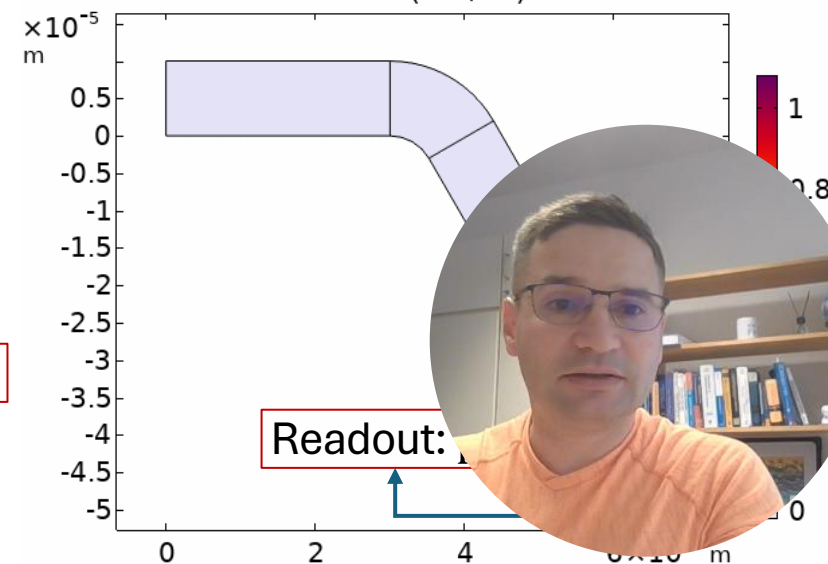
Time=0 s Surface: Molar concentration, $c_{\text{CH}_3\text{COOH}}$ (mol/m^3)



Time=0 s Surface: Molar concentration, c_{NaOH} (mol/m^3)



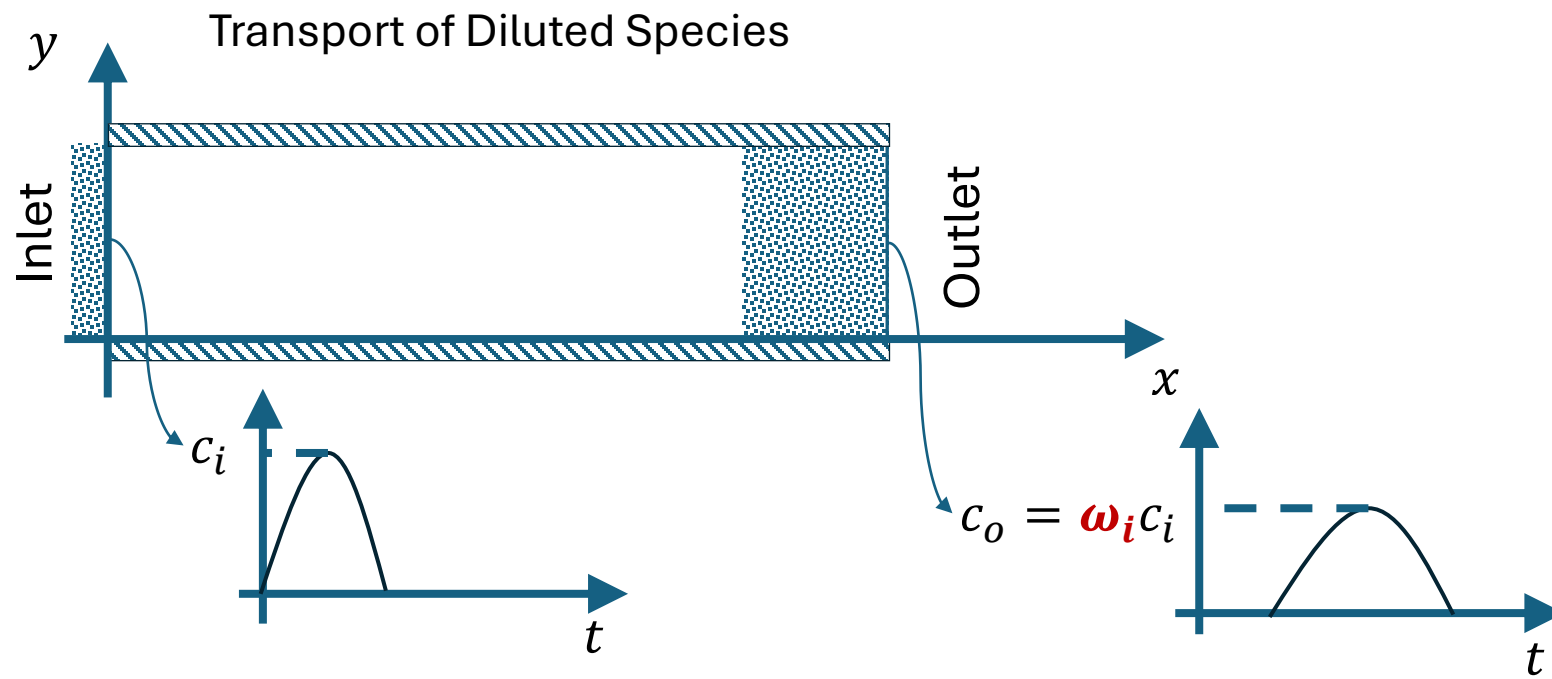
Time=0 s Surface: Molar concentration, $c_{\text{CH}_3\text{COONa}}$ (mol/m^3)



Summary

- Introduction
- Analysis of microfluidic flows
- Synthesis of microfluidic pipes in COMSOL





Analysis

Derivative of the Step-response

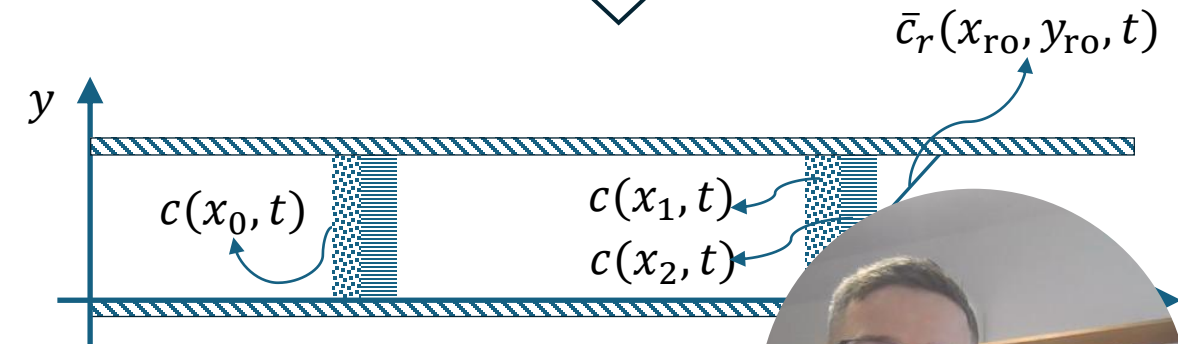
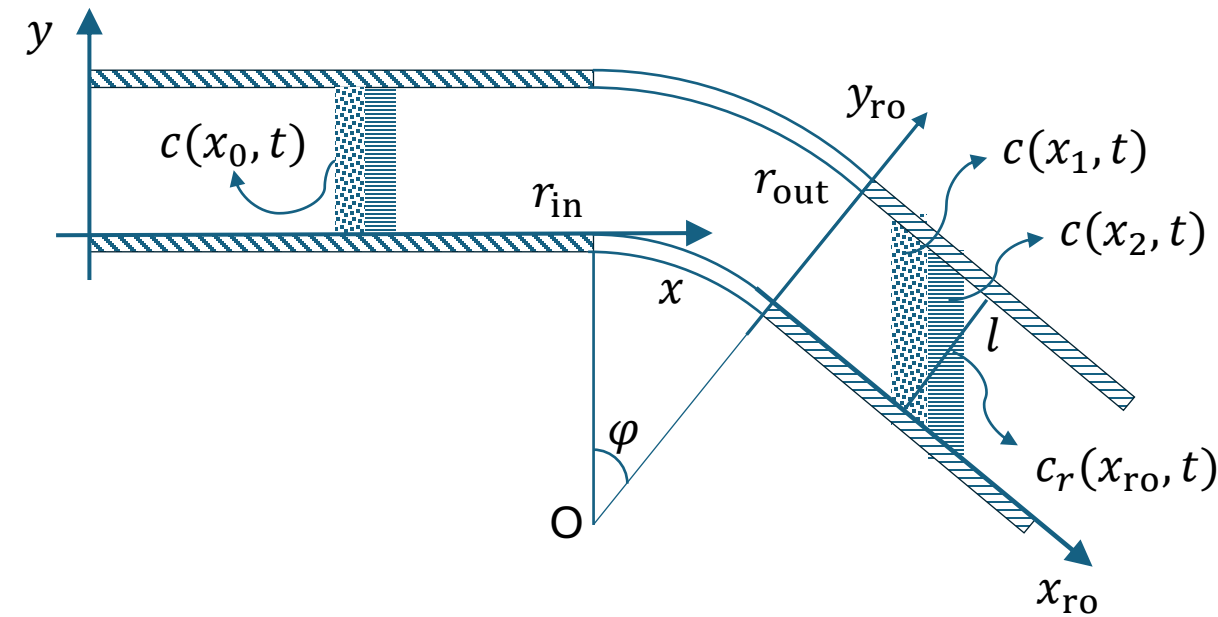
$$h(t) = \left. \frac{\partial}{\partial t} \bar{c}_r(x_{ro}, t) \right|_{C_0=1} \quad (1)$$

$$\bar{c}_r(x_{ro}, t) = \frac{1}{2r} \int_0^{2r} c(x + \varphi y_{ro}, t) dy_{ro} \Big|_{x=x_{ro}+L_{in}+l_{inner}} \quad (2)$$

$$c(x, t) = \frac{1}{2} C_0 \operatorname{erfc} \left(\frac{x - v_{\text{char}} t}{2\sqrt{D_{\text{eff}} t}} \right) \quad (3)$$

$$v_{\text{char}} = \frac{r^2 \Delta p}{3\mu L} \quad (4) \quad D_{\text{eff}} = \frac{r^2 v_{\text{char}}^2}{48D} \quad (5)$$

$$\Delta p = \frac{8\mu L}{\pi r^4} Q \quad (6)$$

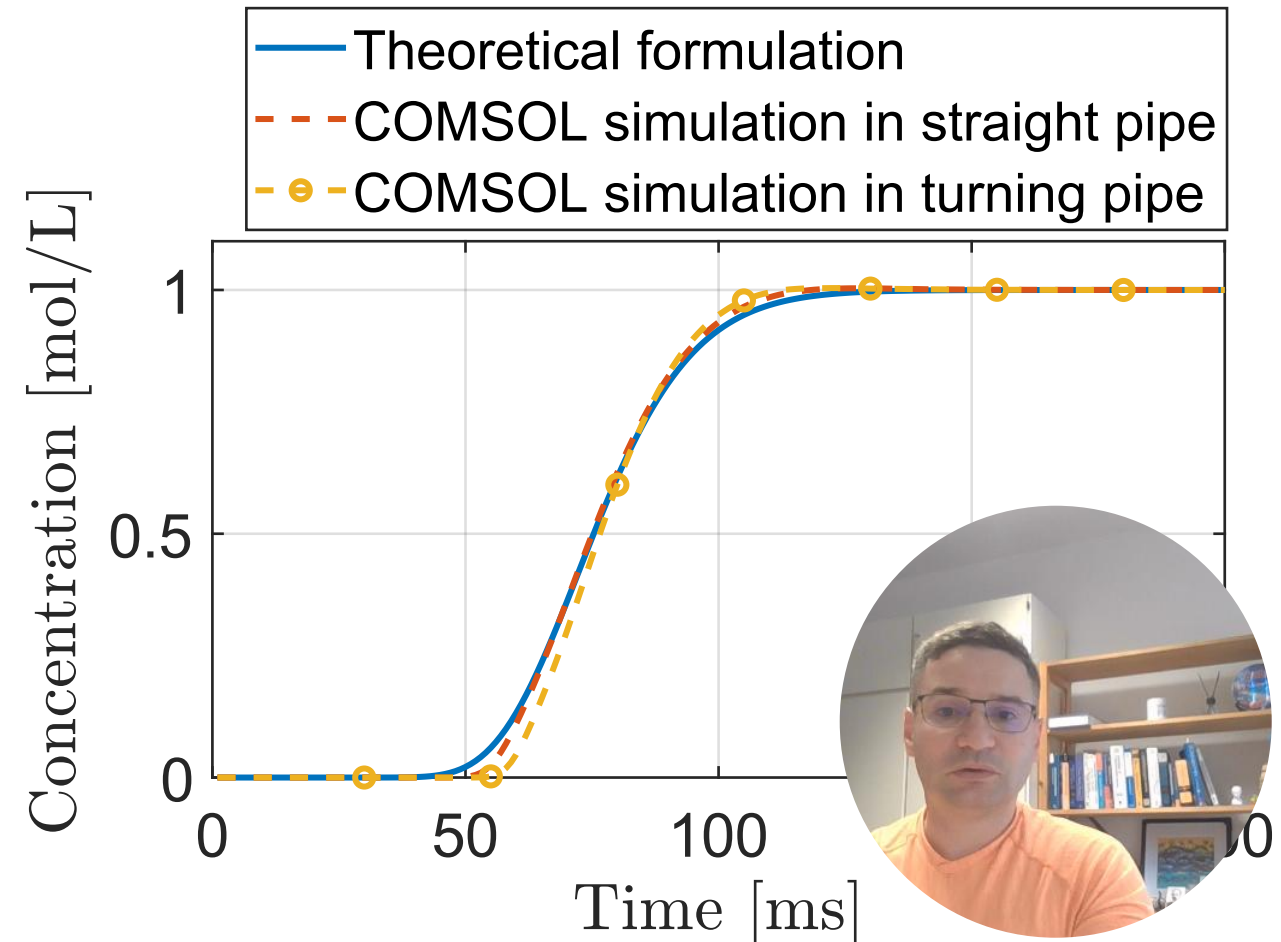
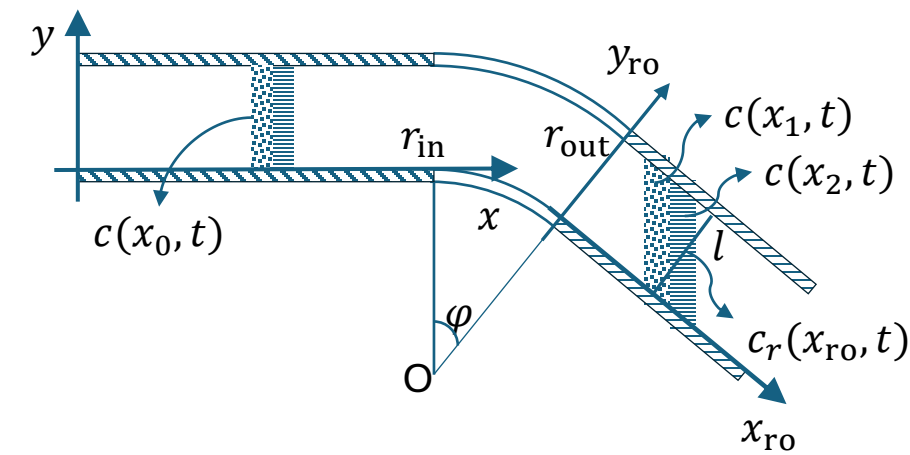


Analysis

Derivative of the Step-response

$$h(t) = \left. \frac{\partial}{\partial t} \bar{c}_r(x_{ro}, t) \right|_{C_0=1} \quad (1)$$

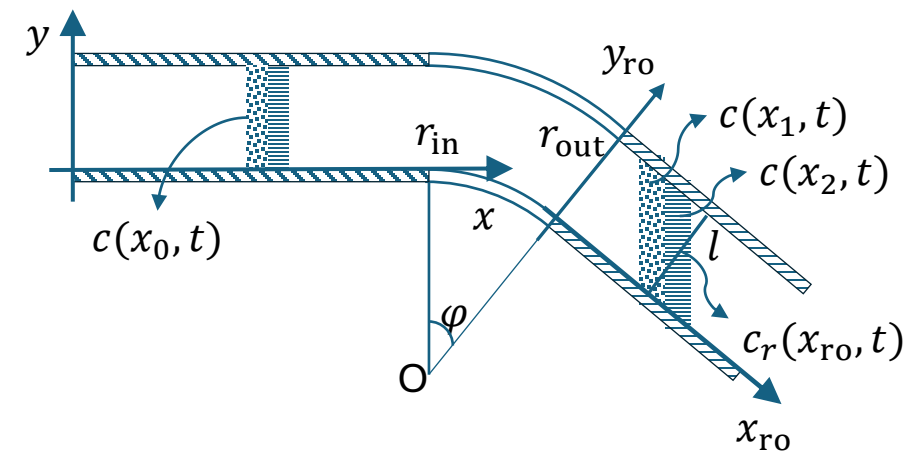
$$\bar{c}_r(x_{ro}, t) = \frac{C_0}{4r} \left[\operatorname{erfc} \left(\frac{x - tv + \varphi y}{2\sqrt{D_{\text{eff}} t}} \right) (x - tv + \varphi y) - 2\sqrt{\frac{D_{\text{eff}} t}{\pi}} e^{-\frac{(x - tv + \varphi y)^2}{4D_{\text{eff}} t}} \right] \bigg|_{y=0}^{y=2r} \quad (2)$$



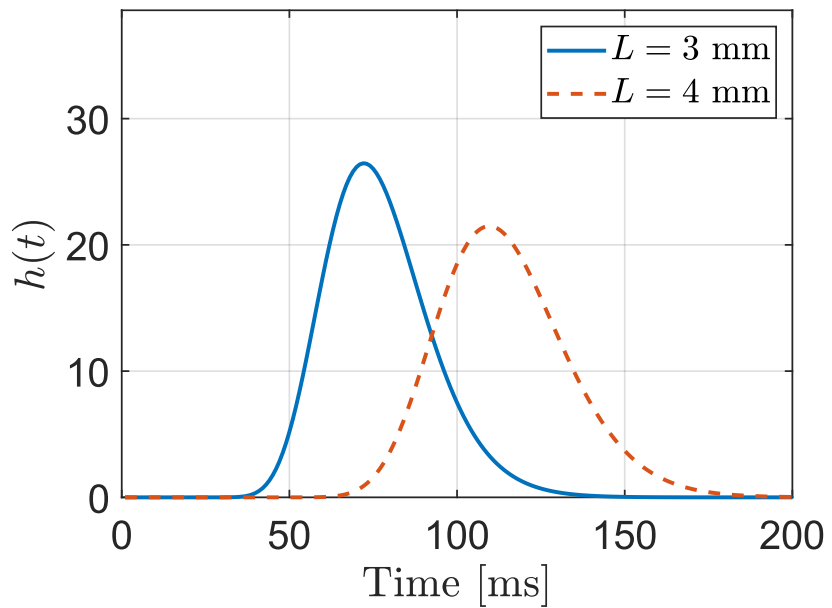
Analysis

Derivative of the Step-response

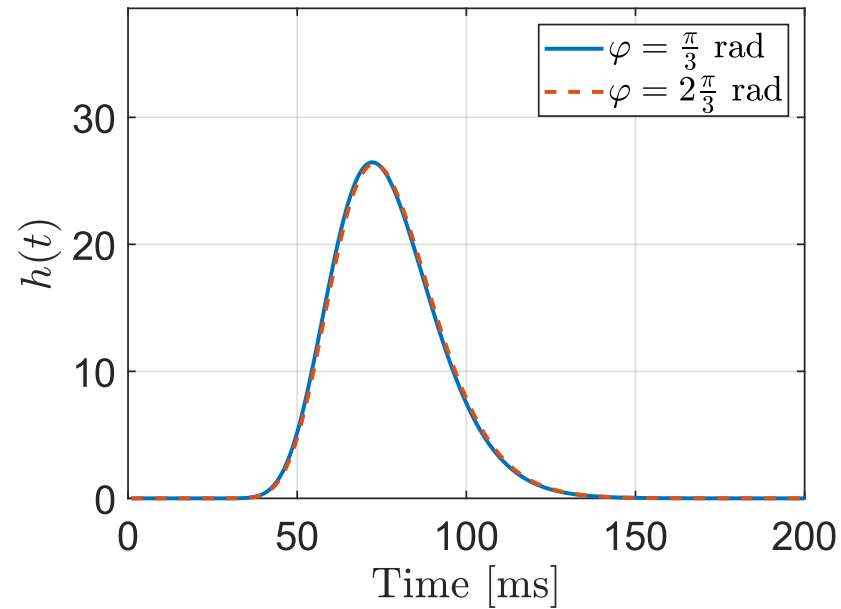
$$h(t) = \left. \frac{\partial}{\partial t} \bar{c}_r(x_{ro}, t) \right|_{C_0=1} \quad (1)$$



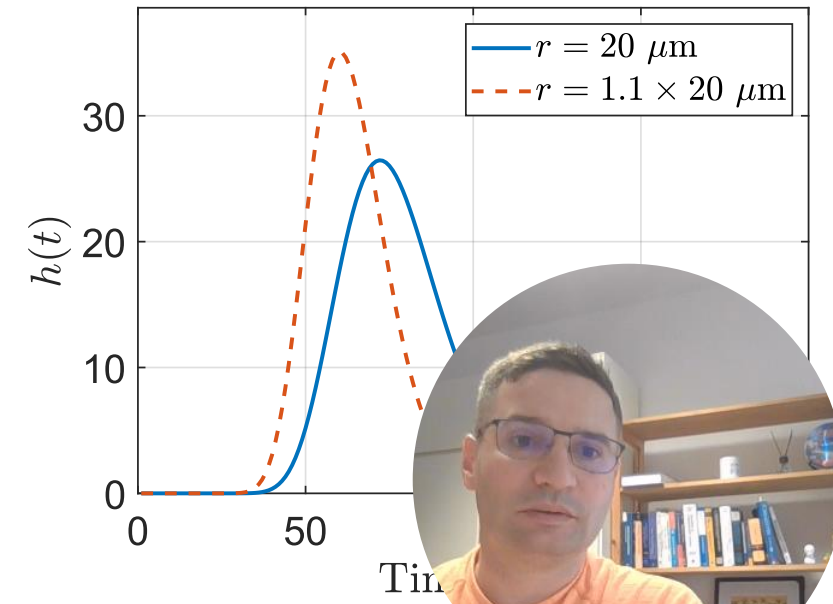
$\varphi = \frac{\pi}{3}$ $r = 20 \mu\text{m}$



$L = 3 \text{ mm}$ $r = 20 \mu\text{m}$



$L = 3 \text{ mm}$ $\varphi = \frac{\pi}{3}$



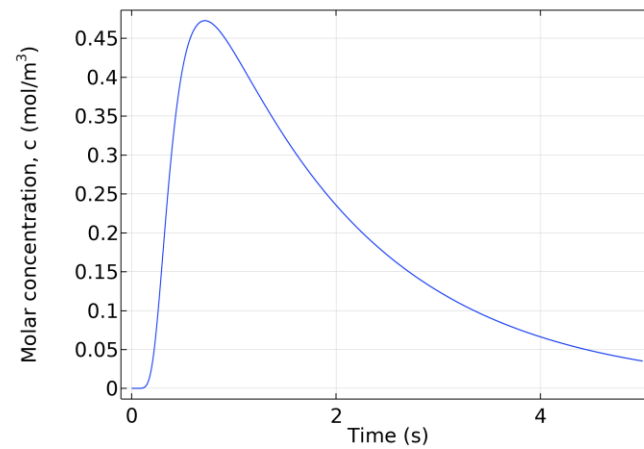
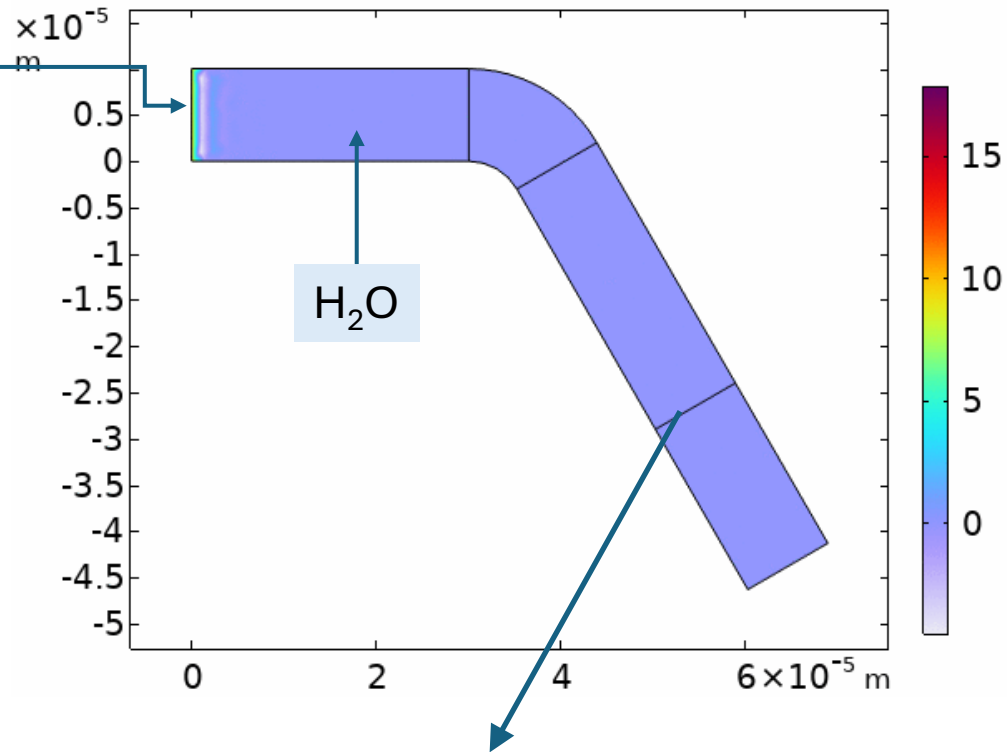
Summary

- Introduction
- Analytic evaluation of advection and diffusion in microfluidic channels
- Synthesis of microfluidic pipes in COMSOL



NaOH

Time=0 s Surface: Molar concentration, $c_{\text{CH}_3\text{COOH}}$
(mol/m^3)



File

New

Model Wizard

Blank Model

File

Select Space Dimension

3D

2D Axisymmetric

2D

Select Physics

Type filter text

Recently Used

Laminar Flow (spf)

Transport of Diluted Species (tds)

Reaction Engineering (re)

AC/DC

Acoustics

Chemical Species Transport

Electric Discharge

Electrochemistry

Fluid Flow

Heat Transfer

Optics

Plasma

Radio Frequency

Semiconductor

Structural Mechanics

Mathematics

Add

Added physics interfaces:

Laminar Flow (spf)

Transport of Diluted Species (tds)

Review Physics Interface

Laminar Flow (spf)

Dependent Variables

Velocity field (m/s):

Velocity field components:

Pressure (Pa):

u

u

v

w

p



Select Physics

Type filter text

Recently Used

- ☒ Laminar Flow (spf)
- ☐ Transport of Diluted Species (tds)
- ☐ Reaction Engineering (re)
- > AC/DC
- > Acoustics
- > Chemical Species Transport
- > Electric Discharge
- > Electrochemistry
- > Fluid Flow
- > Heat Transfer
- > Optics
- > Plasma
- > Radio Frequency
- > Semiconductor
- > Structural Mechanics
- > Mathematics

Laminar Flow

Add

Added physics interfaces:

- ☒ Laminar Flow (spf)
- ☐ Transport of Diluted Species (tds)

Review Physics Interface

Laminar Flow (spf)

Dependent Variables

Velocity field (m/s):

Velocity field components:

Pressure (Pa):

Fluid Velocity Components

(u, v, w) [m/s]

Inlet Pressure

p [Pa]



File

Untitled.mph - COMSOL Multiphysics

Select Physics

Type filter text

Recently Used

Laminar Flow (spf)

Transport of Diluted Species (tds)

Reaction Engineering (re)

AC/DC

Acoustics

Chemical Species Transport

Electric Discharge

Electrochemistry

Fluid Flow

Heat Transfer

Optics

Plasma

Radio Frequency

Semiconductor

Structural Mechanics

Mathematics

Add

Added physics interfaces:

Laminar Flow (spf)

Transport of Diluted Species (tds)

Review Physics Interface

Transport of Diluted Species (tds)

Dependent Variables

Number of species:1

Concentrations (mol/m³):c

+

Transport of Diluted Species

Concentration of Species
 $c \text{ [mol/m}^3\text{]}$



Select Study

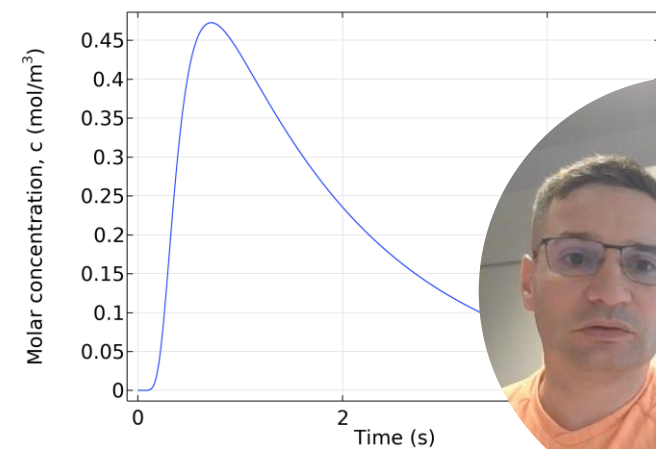
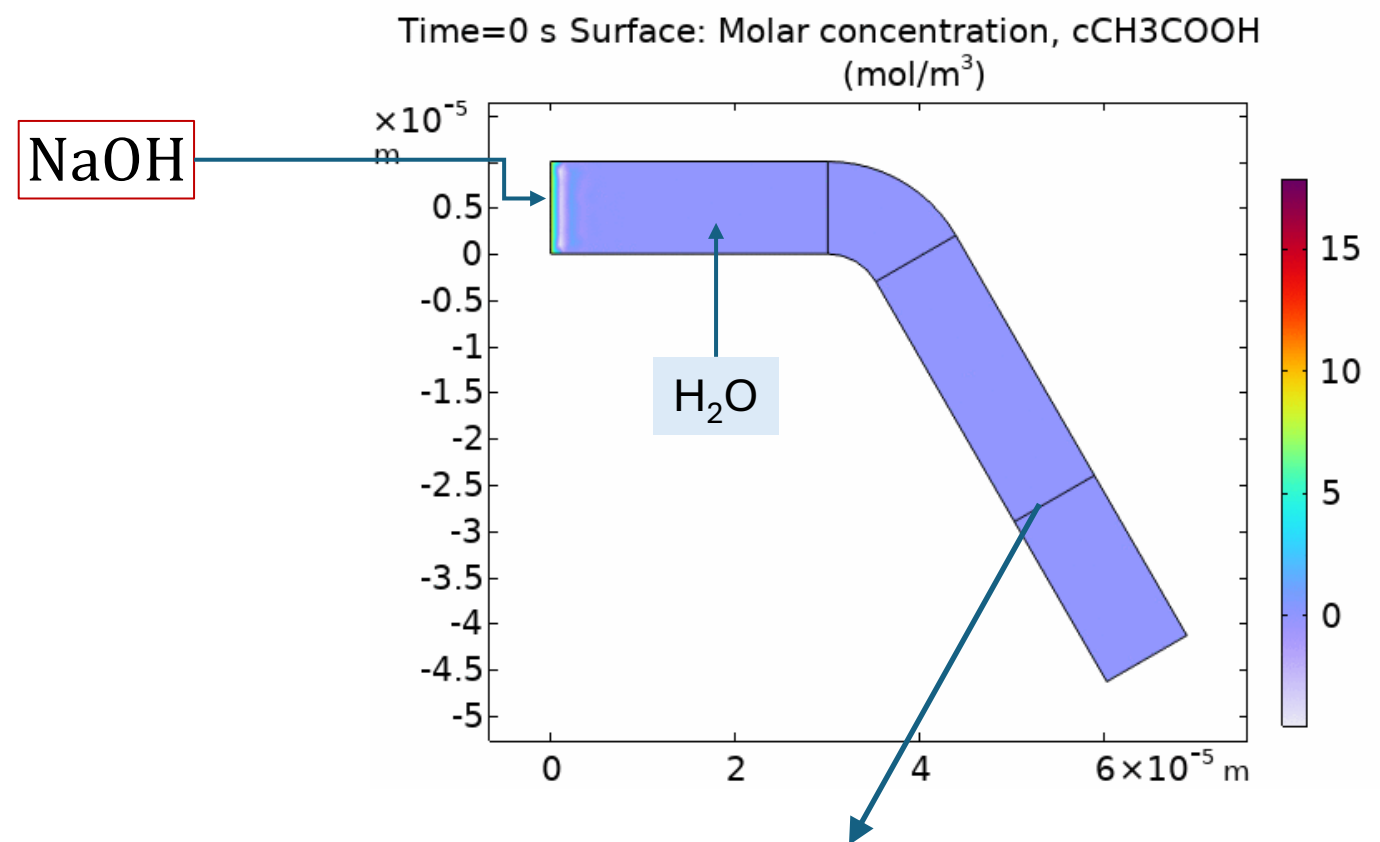
- General Studies
 - Stationary
 - Time Dependent
 - Empty Study

Stationary:

H₂O Flow

Time Dependent:

Transport of Diluted Species NaOH



Application Builder Model Manager Component 1 Add Component Parameters Variables Functions Parameters Case Build All Import LiveLink Part Libraries Add Material Laminar Flow Add Physics Add Mathematics Build Mesh Mesh 1 Compute Study 1 Add Study Select Plot Group Add Plot Group Result Templates Windows Reset Desktop

Model Builder

Type filter text

- Untitled.mph (root)
 - Global Definitions
 - Parameters 1
 - Materials
 - Component 1 (comp 1)
 - Definitions
 - Geometry 1**
 - Materials
 - Laminar Flow (spf)
 - Fluid Properties 1
 - Initial Values 1
 - Wall 1
 - Transport of Diluted Species (tds)
 - Fluid 1
 - No Flux 1
 - Initial Values 1
 - Multiphysics
 - Mesh 1
 - Study 1
 - Step 1: Stationary
 - Results

Settings

Geometry

Build All

Label: Geometry 1

Units

☐ Scale values when changing units

Length unit: m

Angular unit: Degrees

Advanced

Geometry representation: CAD kernel

☒ Design Module Boolean operations

Default repair tolerance: Automatic

☒ Build new operations automatically

☒ Build automatically when leaving geometry

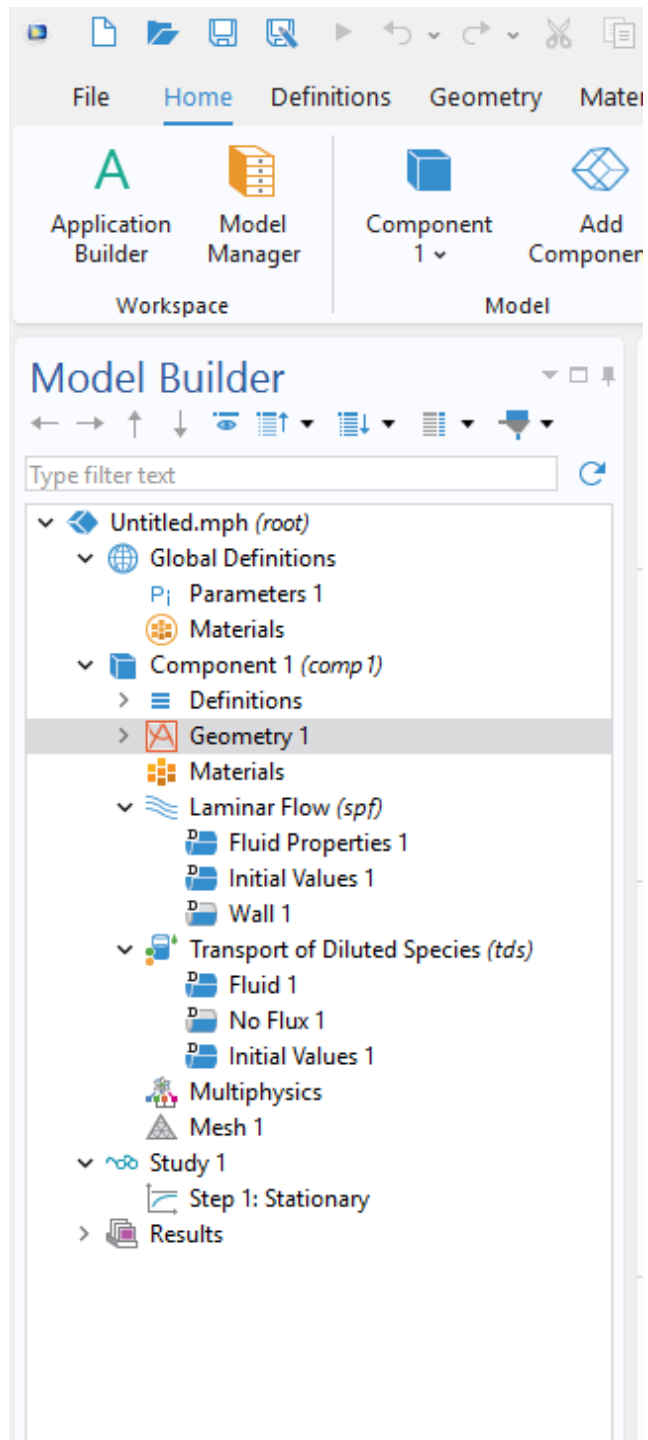
Cleanup

☒ Automatic detection of small details

Graphics

Navigation and visualization tools for the model results.





Components:

- Geometry
- Laminar Flow Physics
- Transport of Diluted Species Physics
- Multiphysics
- Study
- Results



File Home Definitions **Geometry** Materials Physics Mesh Study Results Developer

Build All Import Defeaturing and Repair Cleanup Wizard Virtual Operations Cleanup Block Sphere Cone Cylinder More Primitives Select Work Plane Work Plane Extrude Revolve Sweep Loft Booleans and Partitions Transforms Extract Delete Part Libraries Create Part Load Part Selections Selection List Colors Programming Measure Measurements Clear Sequence

Model Builder

Type filter text

- Untitled.mph (root)
 - Global Definitions
 - Parameters 1
 - Materials
 - Component 1 (comp 1)
 - Definitions
 - Geometry 1**
 - Cylinder 1 (cy1)
 - Form Union (fin)
 - Materials
 - Laminar Flow (spf)
 - Fluid Properties 1
 - Initial Values 1
 - Wall 1
 - Transport of Diluted Species (tds)
 - Fluid 1
 - No Flux 1
 - Initial Values 1
 - Multiphysics
 - Mesh 1
 - Study 1
 - Step 1: Stationary
 - Results

Settings

Cylinder

Build Selected Build All Objects

Label: Cylinder 1

Object Type

Type: Solid

Size and Shape

Radius: $10\text{E-}6$ m

Height: $3\text{E-}5$ m

Position

x: 0 m

y: 0 m

z: 0 m

Axis

Axis type: z-axis

Rotation Angle

Rotation: 0 deg

Coordinate System

Work plane: xy-plane

Layers

Selections of Resulting Entities

☐ Resulting objects selection

Show in physics: Domain selection

Color: None

☐ Create layer selections

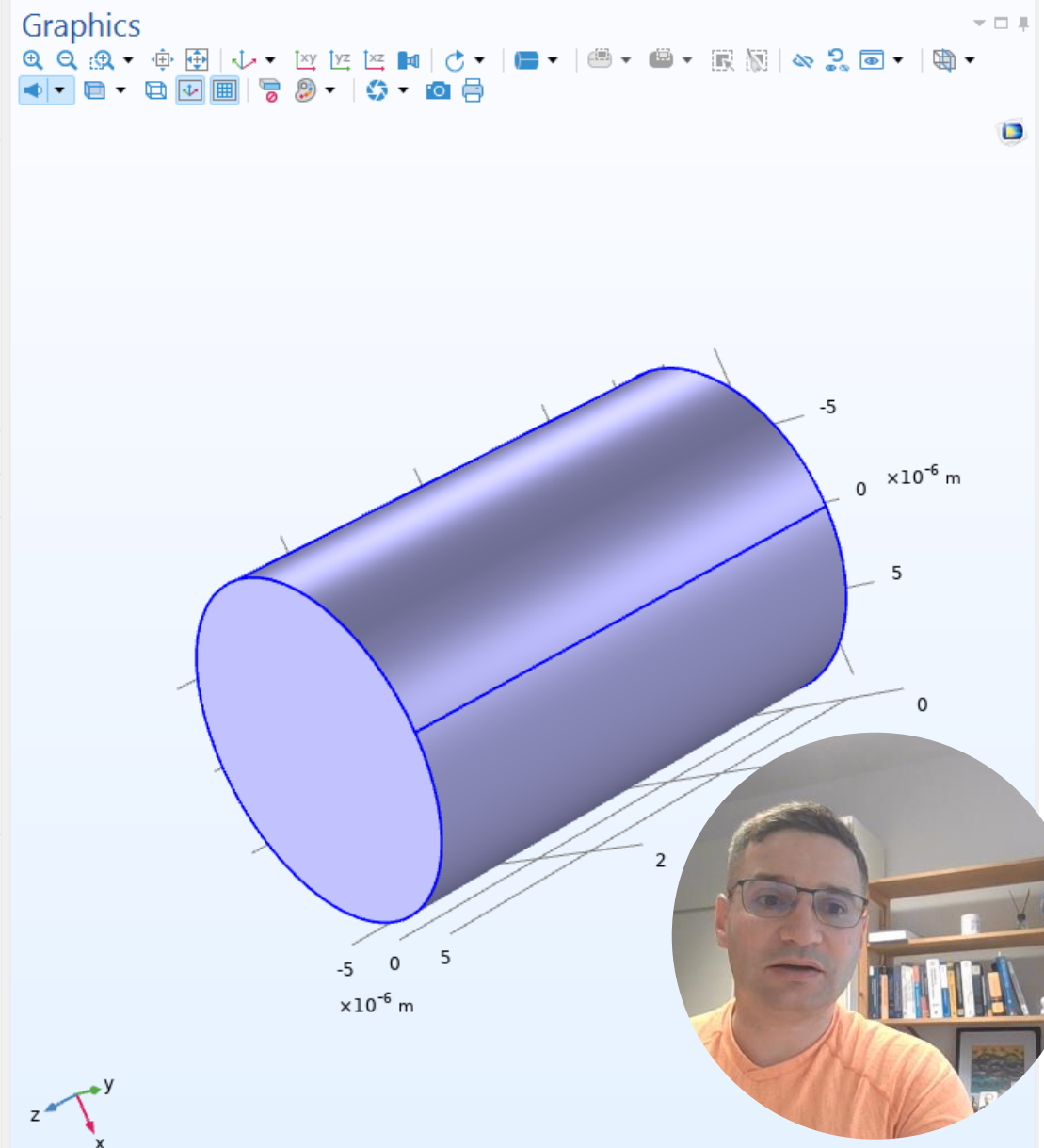
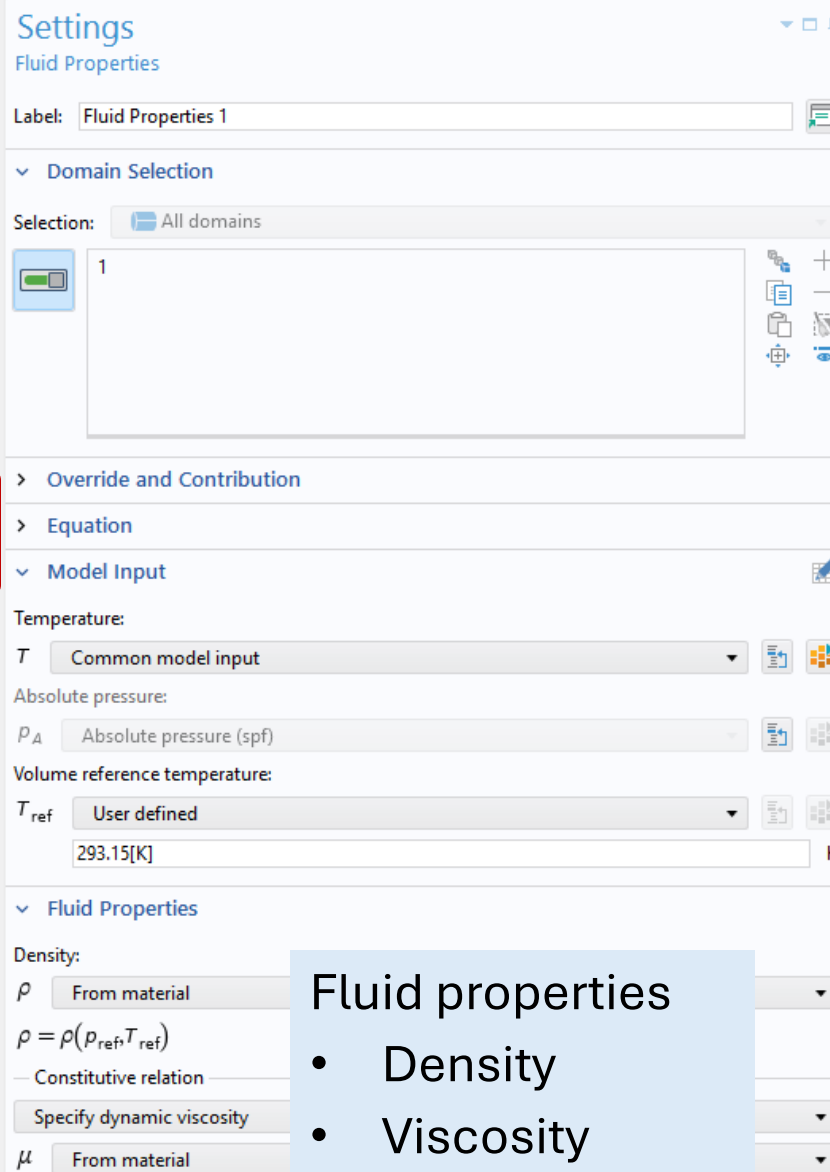
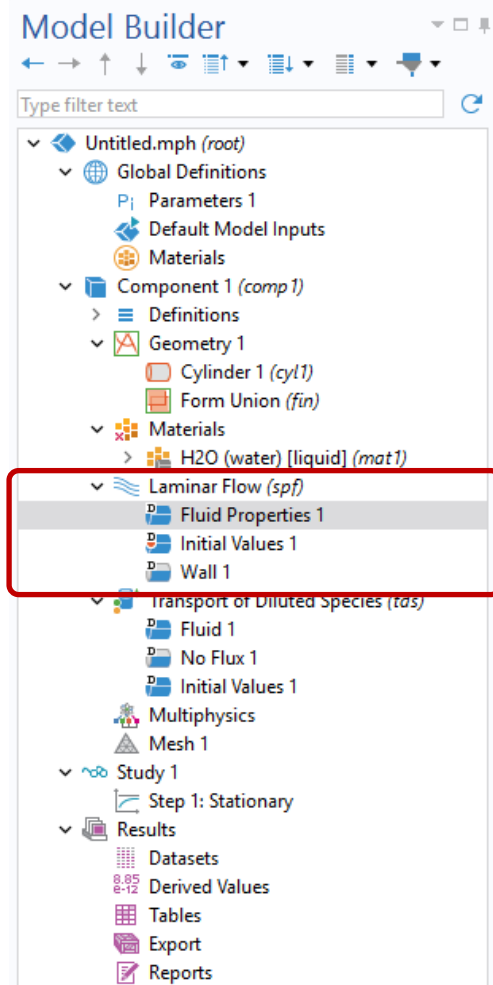
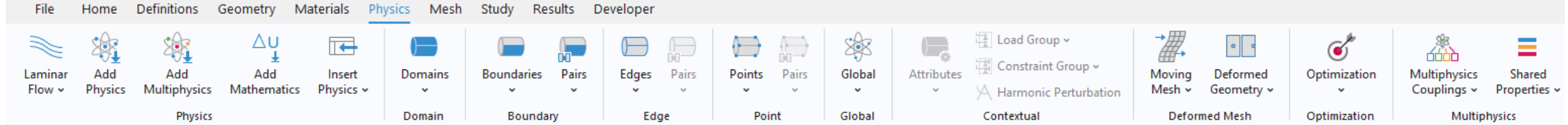
☒ Show in physics

Graphics

Cylinder:

- Radius: $10\text{ }\mu\text{m}$
- Length: $30\text{ }\mu\text{m}$

The 3D model shows a cylinder with a radius of $10\text{ }\mu\text{m}$ and a length of $30\text{ }\mu\text{m}$. The cylinder is centered at the origin of a coordinate system with axes x, y, and z. The x and y axes range from -5 to 5, and the z axis ranges from -5 to 5, all in units of 10^{-6} m. A circular inset in the bottom right corner shows a man with glasses and an orange shirt.



Fluid properties

- Density
- Viscosity

File Home Definitions Geometry Materials **Physics** Mesh Study Results Developer

Laminar Flow Add Physics Add Multiphysics Add Mathematics Insert Physics Domains

Physics Domain

Model Builder

Settings

Inlet

Label: Inlet 1

Boundary Selection

Selection: Manual

Velocity

Normal inflow velocity

Velocity field

U_0 0 m/s

Recently Used

Inlet

Outlet

Laminar Flow

Wall

Symmetry

Periodic Flow Condition

Interior Wall

Flow Control Devices

Fan

Grille

Vacuum Pump

Interior Fan

Screen

Rotating Machinery

Rotating Shaft

Mixing Plane

Fluid Interface Features

Free Surface

Stationary Free Surface

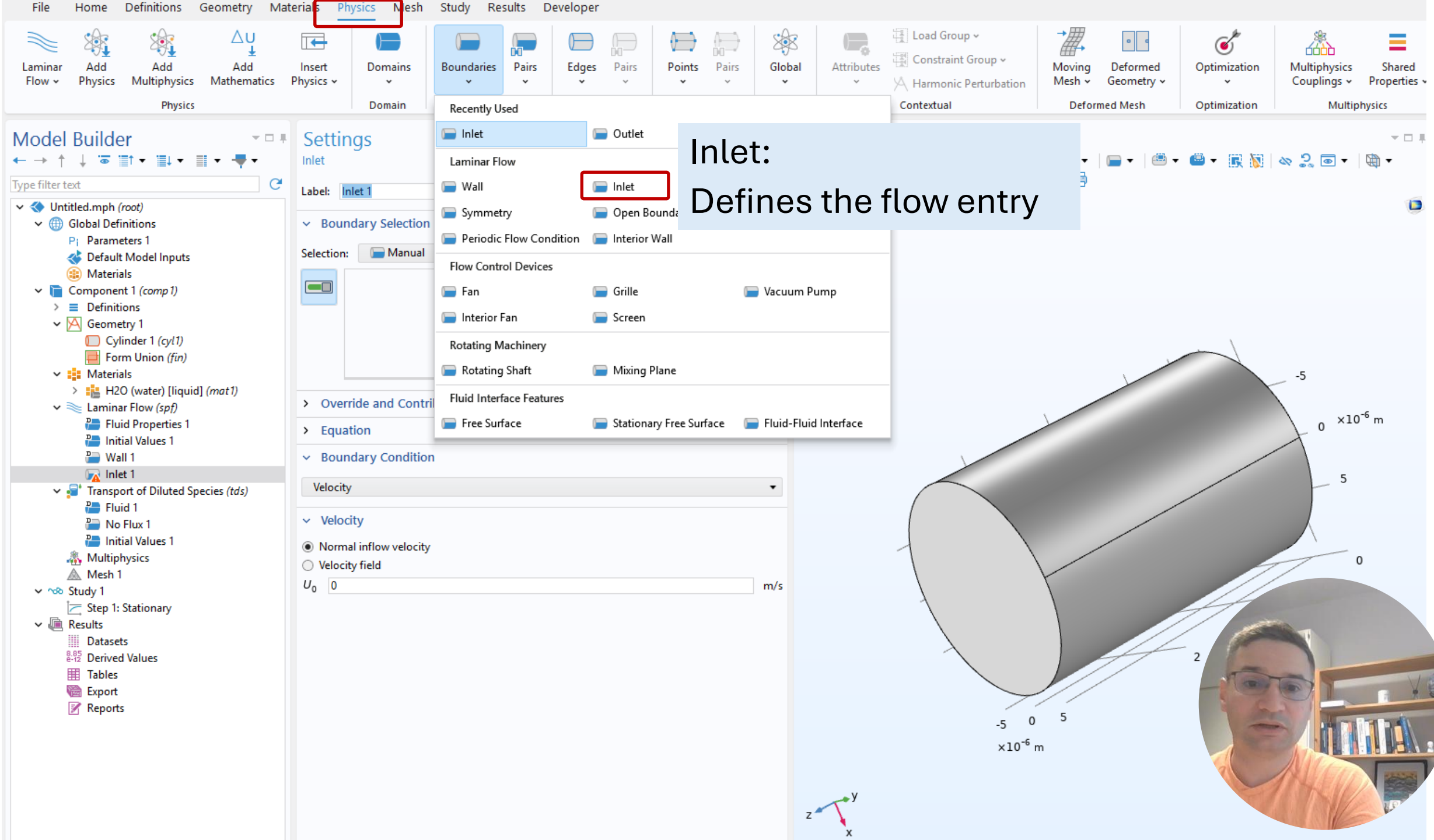
Fluid-Fluid Interface

Inlet: Defines the flow entry

5 0 -5 $\times 10^{-6}$ m

5 0 -5 $\times 10^{-6}$ m

z y x



The image is a screenshot of the COMSOL Multiphysics software interface. The top menu bar includes File, Home, Definitions, Geometry, Materials, **Physics** (highlighted with a red box), Mesh, Study, Results, and Developer. Below the menu is a toolbar with icons for Laminar Flow, Add Physics, Add Multiphysics, Add Mathematics, Insert Physics, and Domains. The main window is divided into three panes. The left pane is the 'Model Builder' showing a tree view of the model structure: 'Untitled.mph (root)' contains 'Global Definitions' (Parameters 1, Default Model Inputs), 'Component 1 (comp 1)' contains 'Definitions', 'Geometry 1' (Cylinder 1 (cyl1), Form Union (fin)), 'Materials' (H2O (water) [liquid] (mat1)), 'Laminar Flow (spf)' (Fluid Properties 1, Initial Values 1, Wall 1, **Inlet 1**), 'Transport of Diluted Species (tds)' (Fluid 1, No Flux 1, Initial Values 1), 'Multiphysics' (Mesh 1), 'Study 1' (Step 1: Stationary), and 'Results' (Datasets, Derived Values, Tables, Export, Reports). The middle pane is the 'Settings' window for the 'Inlet' boundary condition. It shows 'Label: Inlet 1', 'Boundary Selection' (Manual), and 'Velocity' as the boundary condition type. Under 'Velocity', 'Normal inflow velocity' is selected, and the velocity U_0 is set to 0 m/s. The right pane shows a 3D model of a cylinder with coordinate axes (x, y, z). The cylinder is centered at the origin, with x and y axes ranging from -5 to 5 $\times 10^{-6}$ m. A blue text box with the text 'Inlet: Defines the flow entry' is overlaid on the cylinder. A circular inset in the bottom right corner shows a person's face.

Model Builder

- Type filter text
- ▼ Untitled.mph (root)
 - ▼ Global Definitions
 - Parameters 1
 - Default Model Inputs
 - Materials
 - ▼ Component 1 (comp1)
 - Definitions
 - ▼ Geometry 1
 - Cylinder 1 (cyl1)
 - Form Union (fin)
 - Materials
 - H2O (water) [liquid] (mat1)
 - ▼ Laminar Flow (spf)
 - Fluid Properties 1
 - Initial Values 1
 - Wall 1
 - Inlet 1**
 - Transport of Dilute Species (tas)
 - Fluid 1
 - No Flux 1
 - Initial Values 1
 - Multiphysics
 - Mesh 1
 - ▼ Study 1
 - Step 1: Stationary
 - ▼ Results
 - Datasets
 - 8.85 8.12 Derived Values
 - Tables
 - Export
 - Reports

Inlet

Settings

Inlet

Label: Inlet 1

▼ Boundary Selection

Selection: Manual

4

Inlet:
Side number 4

▼ Overview and Connections

▼ Equation

▼ Boundary Condition

Velocity

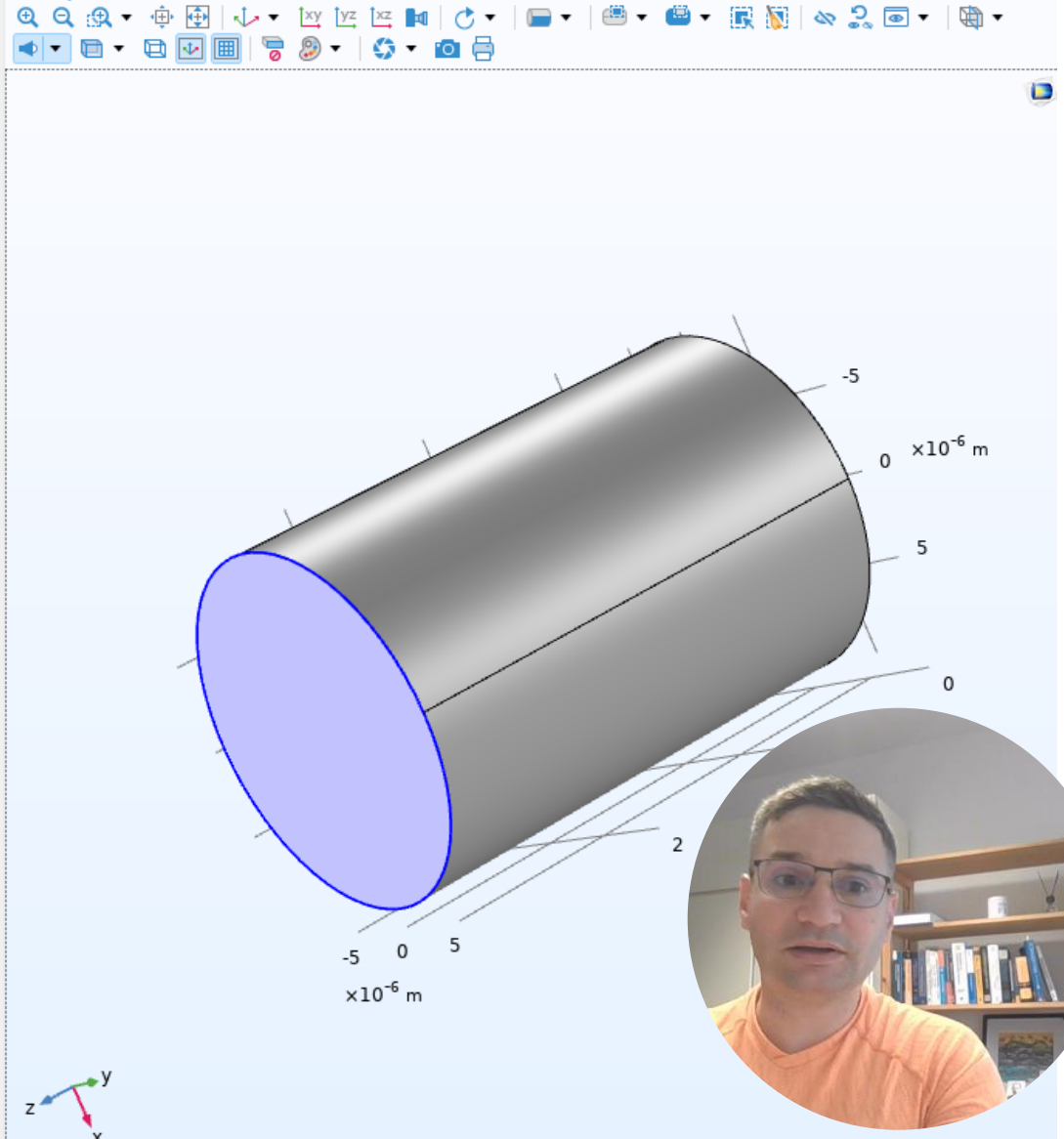
▼ Velocity

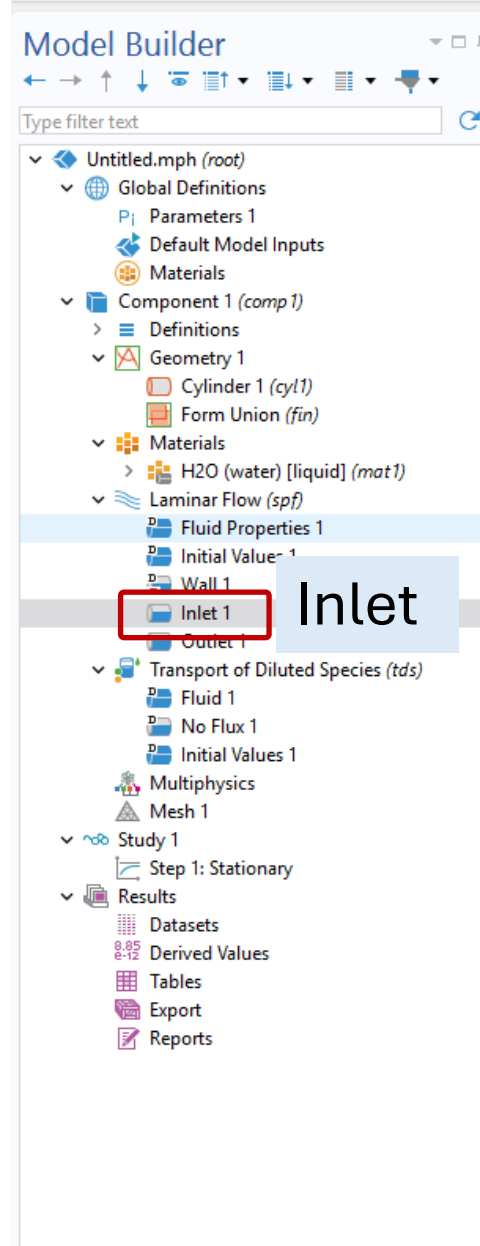
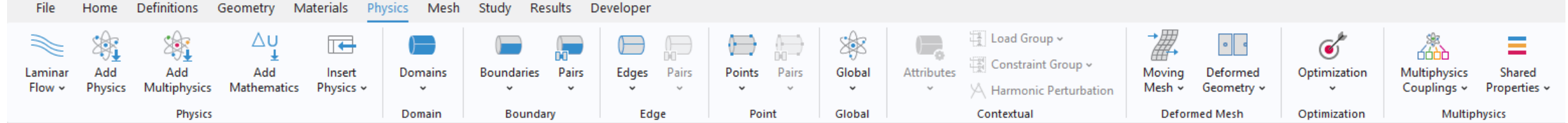
☒ Normal inflow velocity

☐ Velocity field

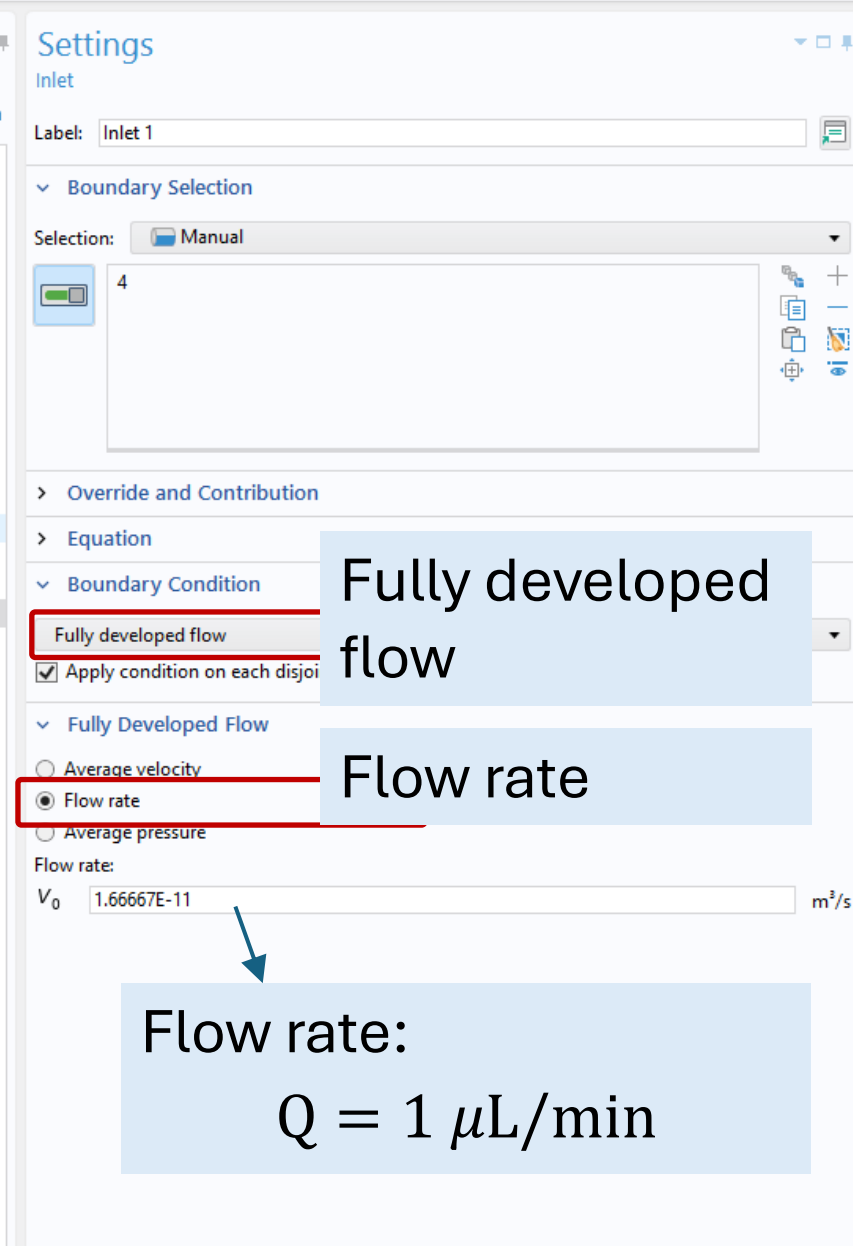
U_0 0 m/s

Graphics





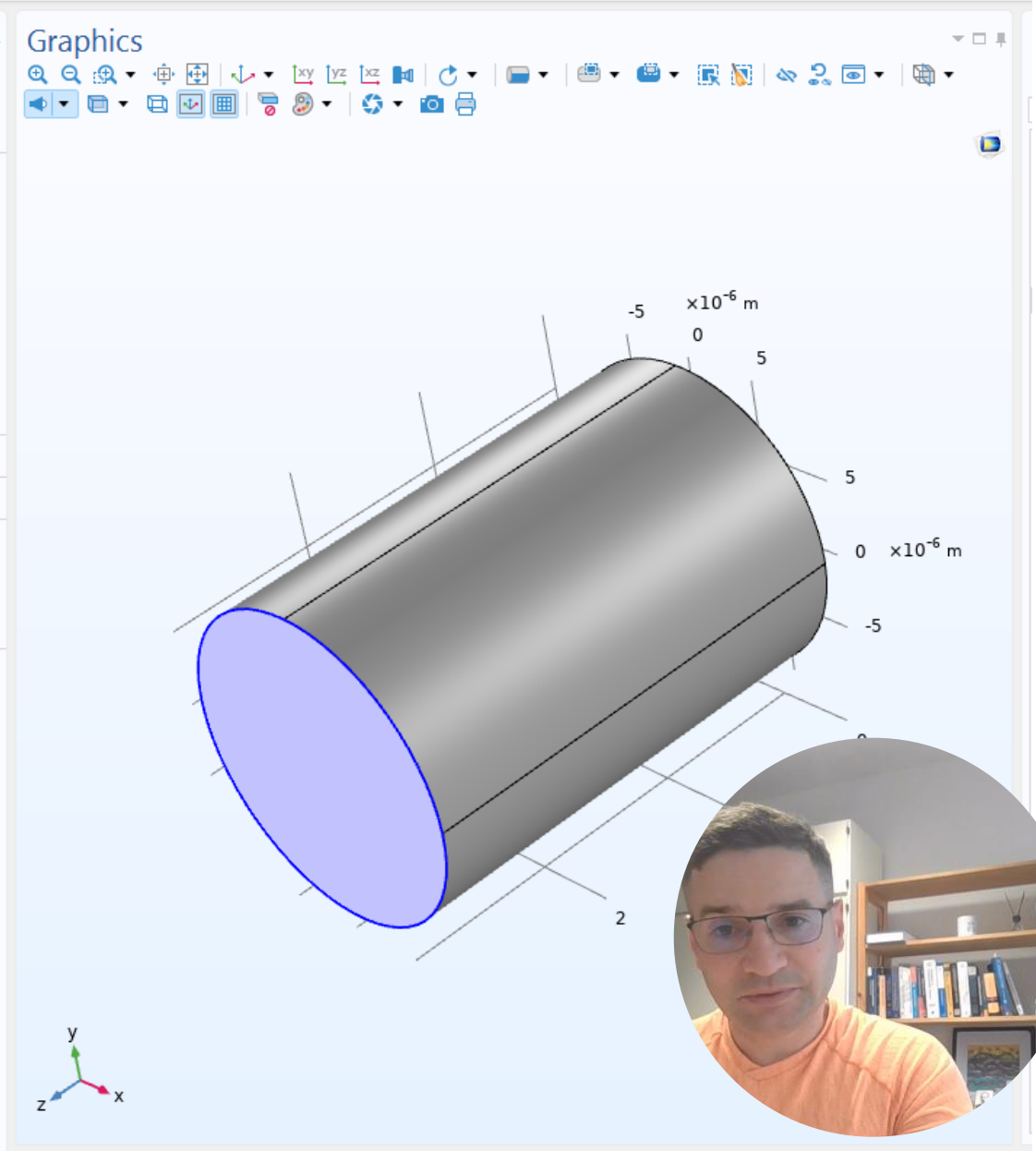
Inlet



Fully developed flow

Flow rate

Flow rate:
 $Q = 1 \mu\text{L}/\text{min}$



File Home Definitions Geometry Materials **Physics** Mesh Study Results Developer

Laminar Flow ▾ Add Physics Add Multiphysics Add Mathematics Insert Physics ▾ Domains ▾ Boundaries ▾ Pairs ▾ Edges ▾ Pairs ▾ Points ▾ Pairs ▾ Global ▾ Attributes ▾ Load Group ▾ Constraint Group ▾ Harmonic Perturbation ▾ Moving Mesh ▾ Deformed Geometry ▾ Optimization ▾ Multiphysics Couplings ▾ Shared Properties

Physics Domain Boundary Edge Point Global Contextual Deformed Mesh Optimization Multiphysics

Model Builder

Type filter text

- Untitled.mph (root)
 - Global Definitions
 - Parameters 1
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 - Materials
 - Component 1 (comp1)
 - Definitions
 - Geometry 1
 - Cylinder 1 (cyl1)
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 - Materials
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 - Laminar Flow (spf)
 - Fluid Properties 1
 - Initial Values 1
 - Wall 1
 - Inlet 1
 - Outlet 1**
 - Transport of Diluted Species (tds)
 - Fluid 1
 - No Flux 1
 - Initial Values 1
 - Multiphysics
 - Mesh 1
 - Study 1
 - Step 1: Stationary
 - Results
 - Datasets
 - 8-95 Derived Values
 - Tables
 - Export
 - Reports

Settings

Outlet

Label: Outlet 1

Boundary Selection

Selection: Manual

3

Outlet:
Side number 3

Override and Contribution

Equation

Boundary Condition

Pressure

Pressure Conditions

Pressure:

Static

P_0 0 Pa

☐ Normal flow

☒ Suppress backflow

Graphics

→ 3 | Up

3D model of a cylinder with a green circular highlight on the side surface, indicating the selected boundary for the outlet condition. The model is shown in a perspective view with coordinate axes (x, y, z) and dimensions in meters (m).



File Home Definitions Geometry Materials **Physics** Mesh Study Results Developer

Transport of Diluted Species Add Physics Add Multiphysics Add Mathematics Insert Physics Domains Boundaries Pairs Edges Pairs Points Pairs Global Attributes Load Group Constraint Group Moving Mesh Deformed Geometry Optimization Multiphysics Couplings

Model Builder

Settings

Fluid

Domain Selection

Selection: All domains

1

Override and Contribution

Equation

Model Input

Temperature:

T Common model input

Coordinate System Selection

Coordinate system: Global coordinate system

Convection

Velocity field:

U Velocity field (spf)

Diffusion

Source:

Material

Material: NaOH [liquid] (mat2)

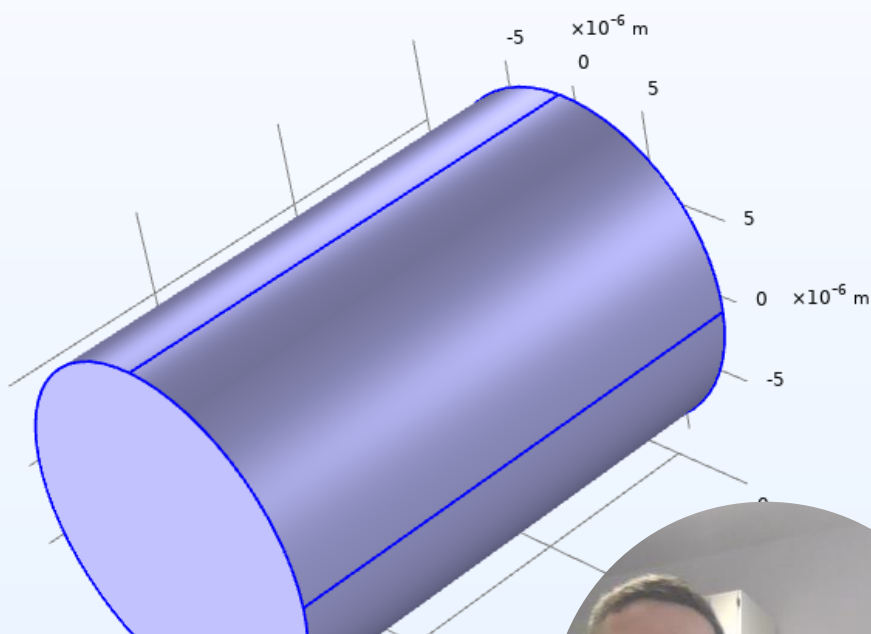
Diffusion coefficient:

D_c User defined

2.125E-9[m^2/s]

Graphics

Convergence Plot 1



Velocity from Laminar Flow Physics

Material: NaOH

Diffusion coefficient: $2.125 \times 10^{-9} \text{ m}^2/\text{s}$



File Home Definitions Geometry Materials **Physics** Mesh Study Results Developer

Transport of Diluted Species Add Physics Add Multiphysics Add Mathematics Insert Physics Domains Boundaries Pairs Edges Pairs Points Pairs Global Attributes Load Group Constraint Group Harmonic Perturbation Moving Mesh Deformed Geometry Optimization Multiphysics Couplings

Model Builder Settings

Type filter text

- Untitled.mph (root)
 - Global Definitions
 - Parameters 1
 - Default Model Inputs
 - Materials
 - Component 1 (comp1)
 - Definitions
 - Rectangle 1 (rect1)
 - Analytic 1 (pulse)
 - Boundary System 1 (sys1)
 - View 1
 - Geometry 1
 - Cylinder 1 (cyl1)
 - Form Union (fin)
 - Materials
 - H2O (water) [liquid] (mat1)
 - NaOH [liquid] (mat2)
 - Basic (def)
 - Thermal expansion (ThermalE)
 - Piecewise (dL)
 - Piecewise 1 (CTE)
 - Laminar Flow (spf)
 - Fluid Properties 1
 - Initial Values 1
 - Wall 1
 - Inlet 1
 - Outlet 1
 - Transport of Diluted Species (tds)
 - Fluid 1
 - No Flux 1
 - Initial values 1
 - Inflow 1**
 - Outflow 1
 - Multiphysics
 - Mesh 1
 - Study 1
 - Step 1: Stationary
 - Solver Configurations
 - Results

Settings

Inflow

Label: Inflow 1

Boundary Selection

Selection: Manual

4

Recently Used

- Outflow
- Inflow
- Transport of Diluted Species
 - Inflow
 - Outflow
 - Concentration
 - Symmetry
 - Periodic Condition
- Flux
- No Flux
- Open Boundary
- Partition Condition
- Flux Discontinuity
- Surface Reactions
 - Surface Reactions
 - Fast Irreversible Surface Reaction
 - Electrode Surface Coupling
- Interior Surfaces
 - Thin Diffusion Barrier
 - Thin Impermeable Barrier

Override and Contribution

Equation

Concentration

$c_{0,c}$ 1 mol/m³

Boundary Condition Type

Concentration constraint

Constant concentration with time

$c = 1 \text{ mol/m}^3$

5 0 $\times 10^{-6} \text{ m}$ -5 0

-5 0 5 $\times 10^{-6} \text{ m}$

y x z

Model Builder

Type filter text

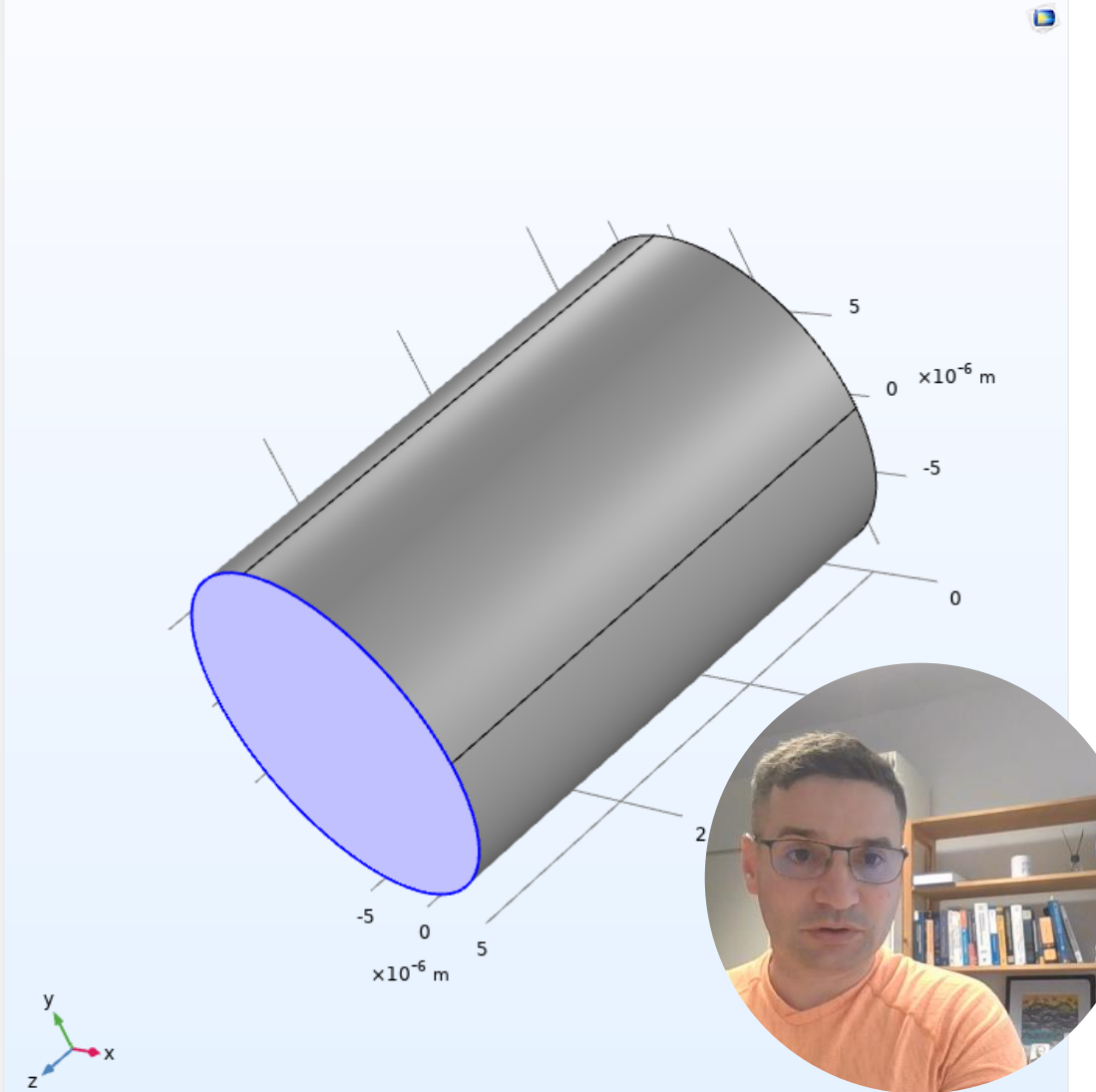
- Untitled.mph (root)
 - Global Definitions
 - Parameters 1
 - Default Model Inputs
 - Materials
 - Component 1 (comp 1)**

- Functions
 - Analytic
 - Interpolation
 - Piecewise
 - Gaussian Pulse
 - Ramp
 - Rectangle**
 - Step
 - Triangle
 - Waveform
 - Normal Distribution
 - Random
 - Elevation (DEM)
 - Image
 - Least-Squares Fit
 - Gaussian Process
 - Polynomial Chaos Expansion
 - Deep Neural Network
 - Partial Fraction Fit
 - Function Switch

Add a Rectangle function

Graphics

Convergence Plot 1



Model Builder

Type filter text

- Untitled.mph (root)
 - Global Definitions
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 - Default Model Inputs
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 - Definitions
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 - Initial Values 1
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 - Outflow 1
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 - Mesh 1
 - Study 1
 - Step 1: Stationary
 - Solver Configurations
 - Results
 - Datasets
 - Derived Values

Settings

Rectangle

Plot Create Plot

Label: Rectangle 1

Function name: rect1

Parameters

Lower limit: 0

Upper limit: 1

Baseline: 0

Amplitude: 1

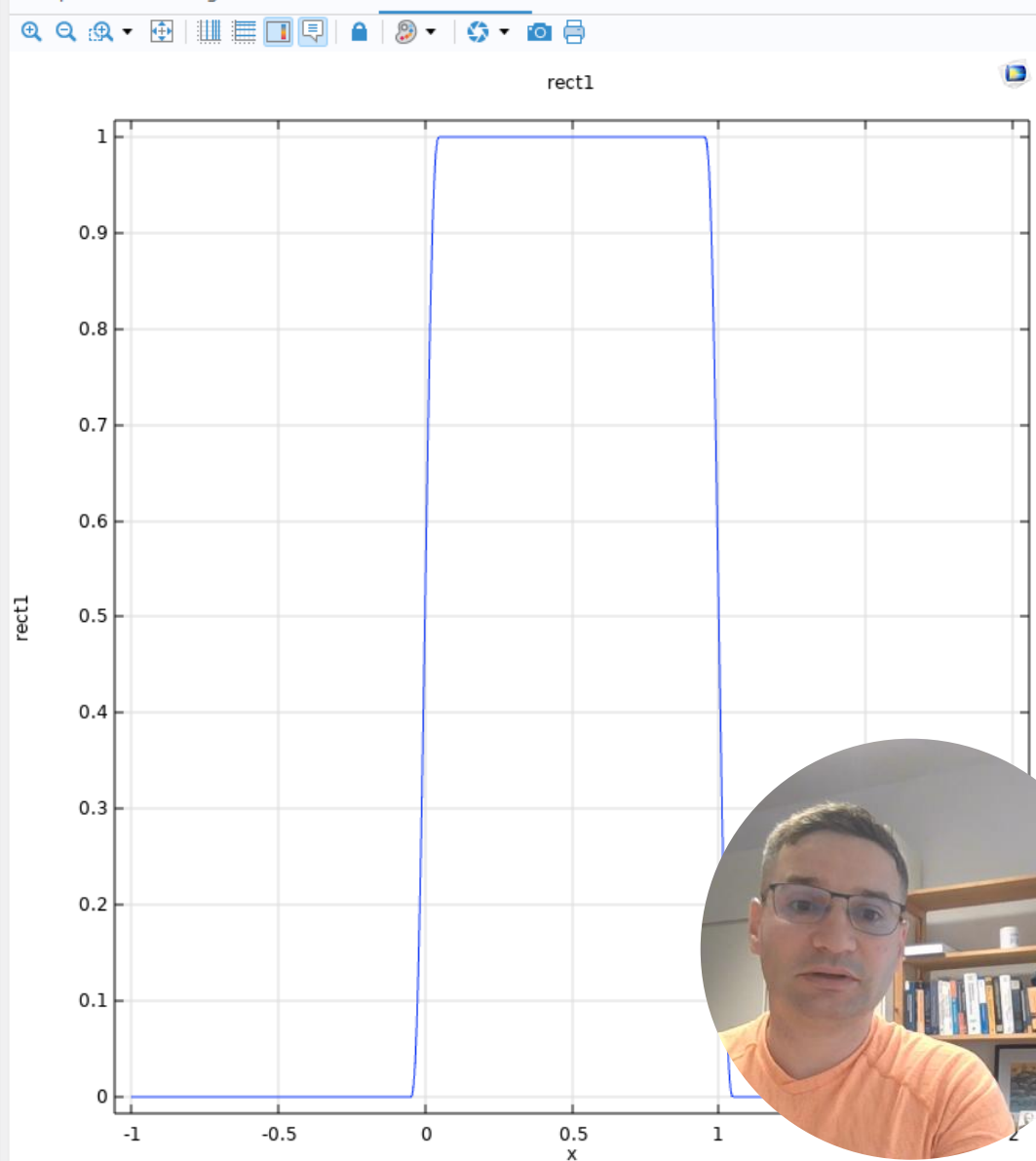
Smoothing

Size of transition zone: ☒ 0.1

Number of continuous derivatives: 2

Rectangle function
Amplitude: 1
Duration: 1

Graphics



Model Builder

Type filter text

- Untitled.mph (root)
 - Global Definitions
 - Parameters 1
 - Default Model Inputs
 - Materials
 - Component 1 (comp 1)
 - Definitions
 - Rect
 - Ana
 - Bou
 - View
 - Geomet
 - Cylind
 - For
 - Material
 - H2O
 - NaC
 - Laminar
 - Fluid
 - Initi
 - Wall
 - Inlet
 - Outl
 - Transpo
 - Fluid
 - No f
 - Initi
 - Inflc
 - Outl

Multiphysics Mesh 1

Settings

Definitions

Add an Analytic function

- Variables
- Functions**
 - Analytic**
 - Interpolation
 - Piecewise
 - Gaussian Pulse
 - Ramp
 - Rectangle
 - Step
 - Triangle
 - Waveform
 - Normal Distribution
 - Random
 - Elevation (DEM)
 - Image
 - Least-Squares Fit
 - Gaussian Process
 - Polynomial Chaos Expansion
 - Deep Neural Network
 - Partial Fraction Fit
 - Function Switch
- Selections
- Probes
- Physics Utilities
- Nonlocal Couplings
- Pairs
- Coordinate Systems
- Infinite Element Domain
- Perfectly Matched Layer
- Shared Properties
- Control Variables
- Surrogate Model Sampling
- View
- Show More Options...
- Node Group
- Group by Type
- Update Probes
- Help

Graphics

Convergence Plot 1 x Function Plot x

5 0 $\times 10^{-6}$ m

-5

-5 0 5 $\times 10^{-6}$ m

Model Builder

Type filter text

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 - Piecewise (dL)
 - Piecewise 1 (CTE)
 - Laminar Flow (spf)
 - Fluid Properties 1
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 - Wall 1
 - Inlet 1
 - Outlet 1
 - Transport of Diluted Species (tds)
 - Fluid 1
 - No Flux 1
 - Initial Values 1
 - Inflow 1
 - Outflow 1
 - Multiphysics
 - Mesh 1
- Study 1
 - Step 1: Stationary
 - Solver Configurations
- Results
 - Datasets

Settings

Analytic

Plot Create Plot

Label: Analytic 1

Function name: pulse

Definition

Expression: rect1(t)

Arguments: t

Derivatives: Automatic

Periodic Extension

Units

Function:

Argument	Unit
t	s

Local Parameters

Advanced

Plot Parameters

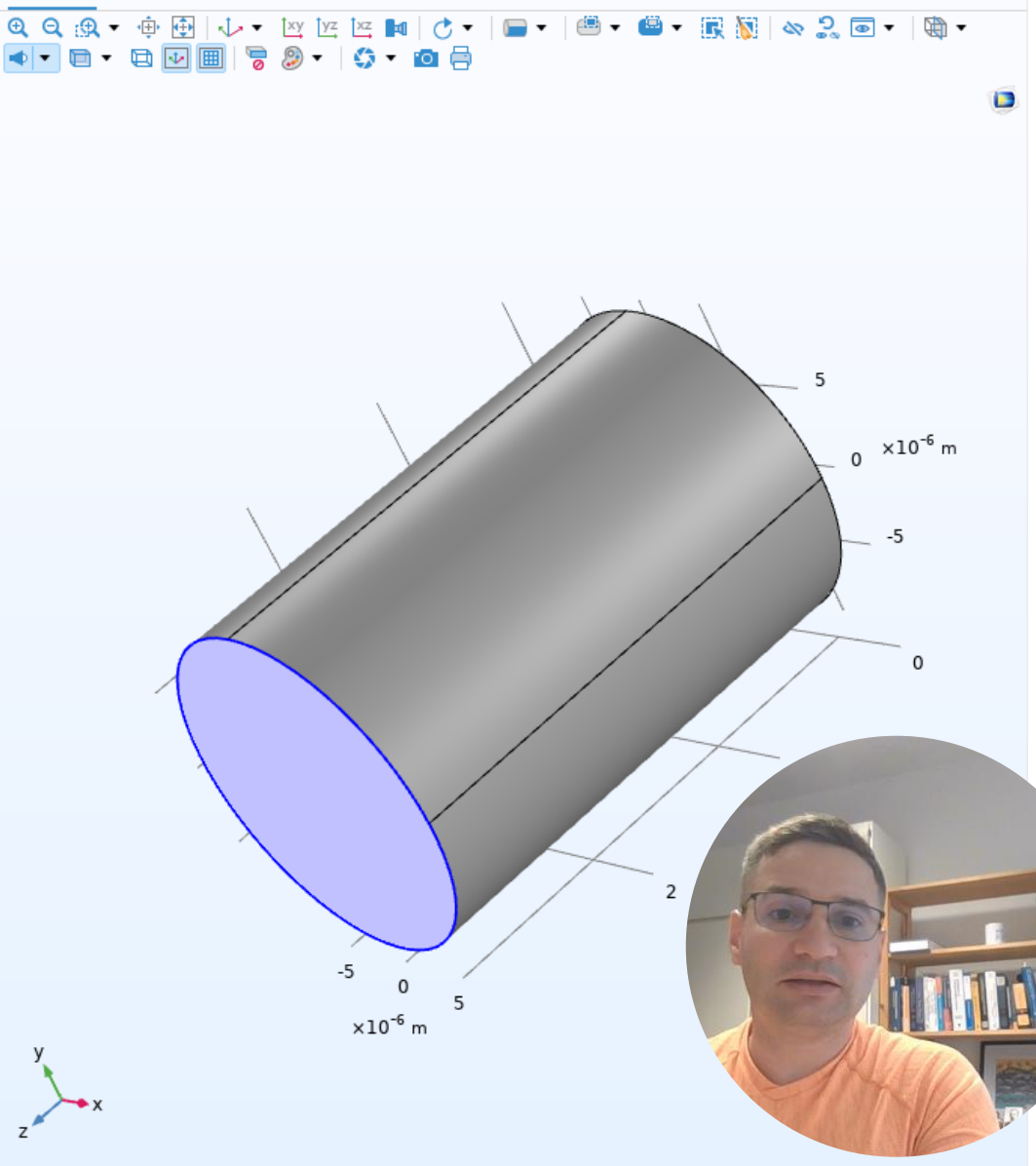
Plot	Argument	Lower limit	Upper limit	Fixed value	Unit
☒	t	0	1	0	s

Analytic function

- Named as „pulse“
- Calls to rect1(t)

Graphics

Convergence Plot 1 × Function Plot ×



Model Builder

Type filter text

- Untitled.mph (root)
 - Global Definitions
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 - Default Model Inputs
 - Materials
 - Component 1 (comp 1)
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 - Rectangle 1 (rect 1)
 - Analytic 1 (pulse)
 - Boundary System 1 (sys 1)
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 - Fluid Properties 1
 - Initial Values 1
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 - Inlet 1
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 - Outflow 1
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 - Step 1: Stationary
 - Solver Configurations
 - Results
 - Datasets

Settings

Inflow

Label: Inflow 1

Boundary Selection

Selection: Manual

4

Override and Contribution

Equation

Concentration

$c_{0,c}$ $10 \cdot \text{pulse}(t)$ mol/m³

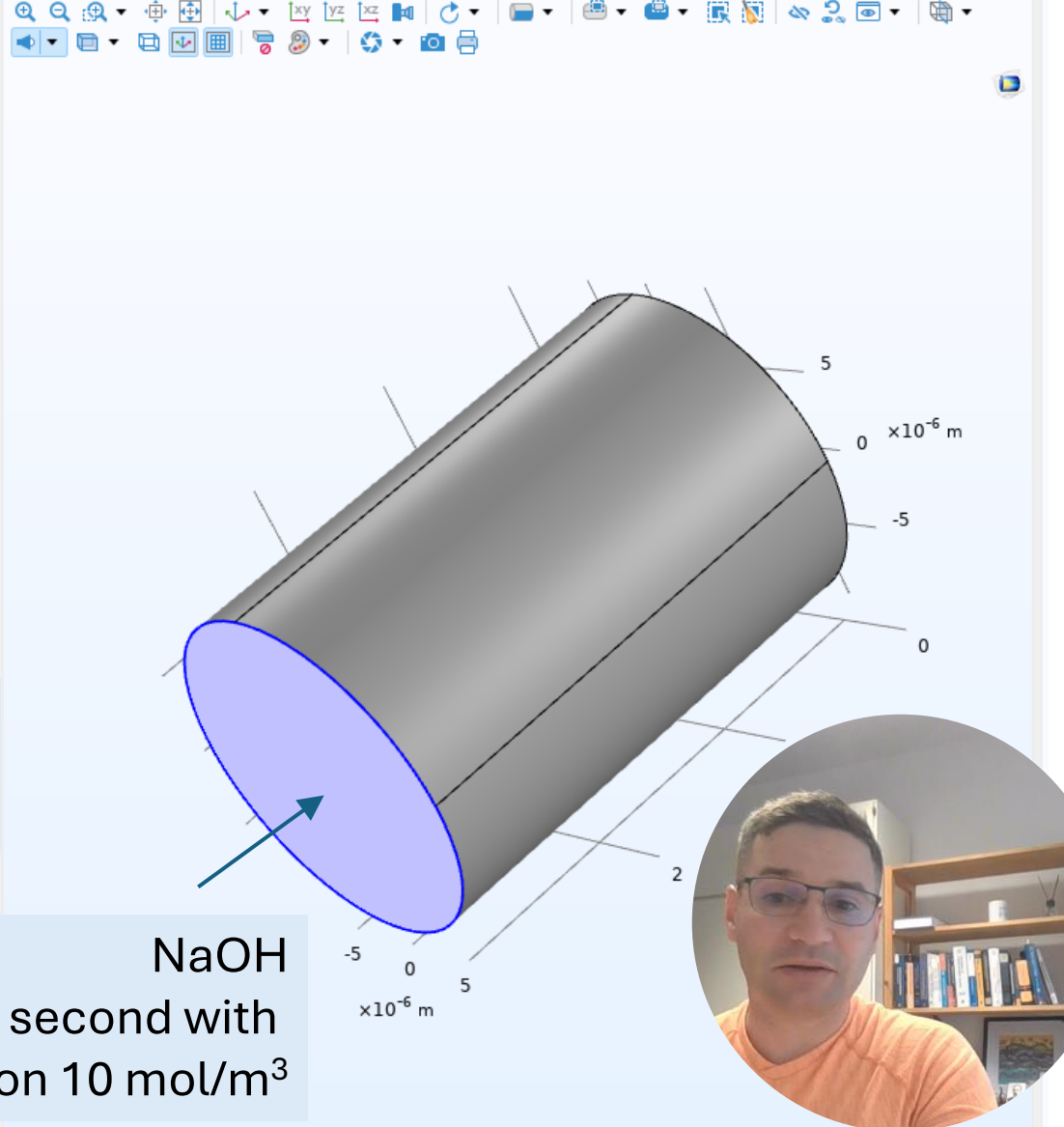
Boundary Condition Type

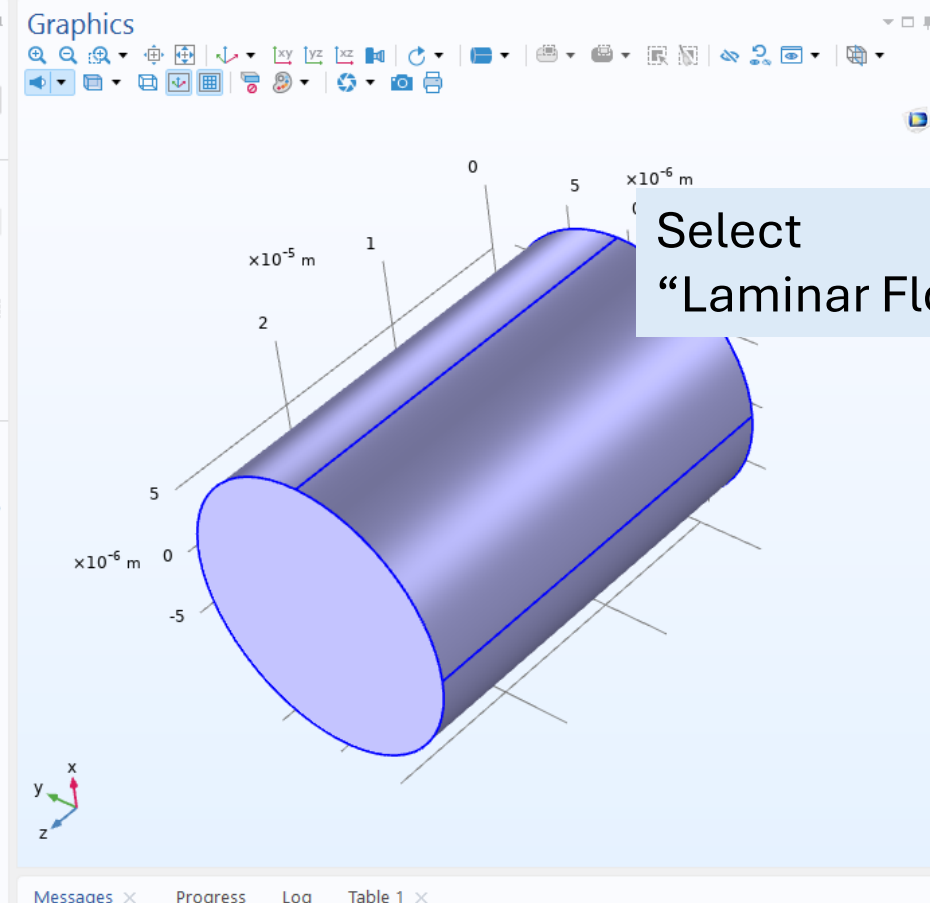
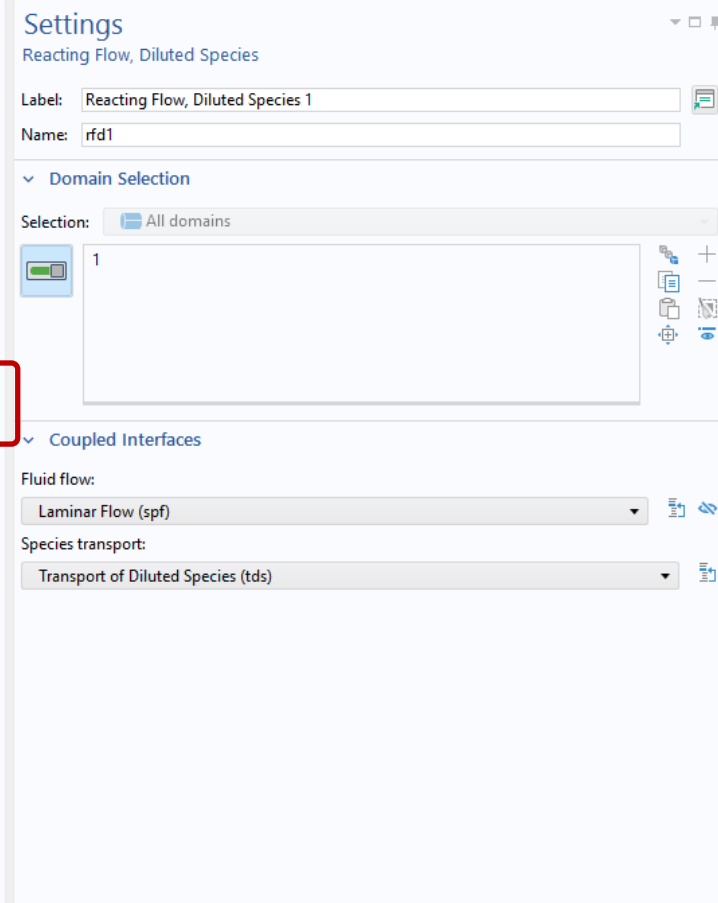
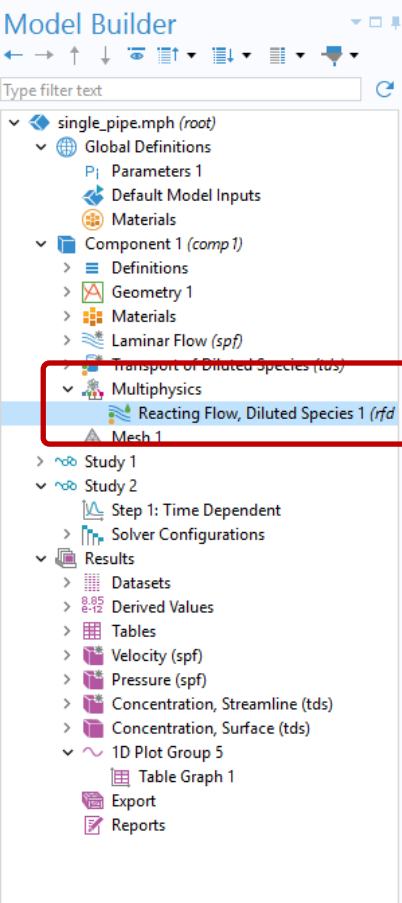
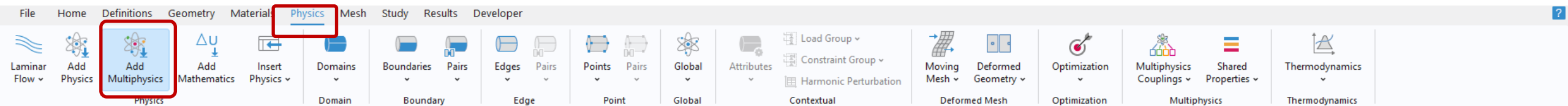
Concentration constraint

Call to the „pulse“ function

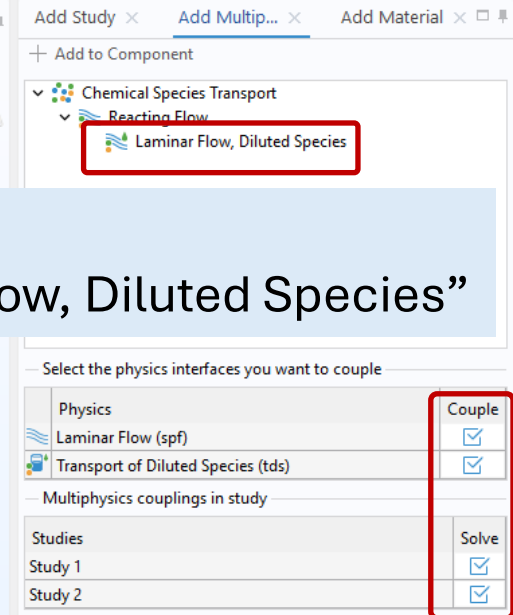
- $10 \cdot \text{pulse}(t)$

Graphics





Select
"Laminar Flow, Diluted Species"



File Home Definitions Geometry Materials Physics Mesh **Study** Results Developer

Compute Study 2 **Add Study** Continue Update Solution Get Initial Value

Show Default Solver Show Default Plots Reset Default Plots

Study Steps Parametric Sweep Function Sweep Material Sweep Combine Solutions Study Reference Optimization Uncertainty Quantification Surrogate Model Training Create Solution Copy Statistics Clear Solutions Clear All Solutions Cosimulation for Simulink

Solver Study Step Operations Evaluate Clear External Interface

Model Builder

Type filter text

- single_pipe.mph (root)
 - Global Definitions
 - Parameters 1
 - Default Model Inputs
 - Materials
 - Component 1 (comp1)
 - Definitions
 - Geometry 1
 - Materials
 - Laminar Flow (spf)
 - Transport of Diluted Species (tds)
 - Multiphysics
 - Reacting Flow, Diluted Species 1 (rfd)
 - Mesh 1
 - Study 1
 - Study 2**
 - Step 1: Time Dependent**
 - Solver Configurations
 - Results
 - Datasets
 - Derived Values
 - Tables
 - Velocity (spf)
 - Pressure (spf)
 - Concentration, Streamline (tds)
 - Concentration, Surface (tds)
 - 1D Plot Group 5
 - Table Graph 1
 - Export
 - Reports

Settings

Time Dependent

Compute Update Solution

Label: Time Dependent

Study Settings

Time unit: s

Output times: range(0,0.1,5) s

Tolerance: Physics controlled

Results While Solving

Physics and Variables Selection

☐ Modify model configuration for study step

Node	Solve for	Equation form
Component 1 (comp1)	☒	
Laminar Flow (spf)	☐	Automatic (Stationary)
Transport of Diluted Species (tds)	☒	Automatic (Time dependent)
Multiphysics		
Reacting Flow, Diluted Species 1 (...)	☒	Automatic (Time dependent)

Values of Dependent Variables

Initial values of variables solved for

Settings: User controlled

Method: Initial expression

Study: Study 1, Stationary

Solution: Current (Solution 1 (sol1))

Selection: Automatic (single solution)

Values of variables not solved for

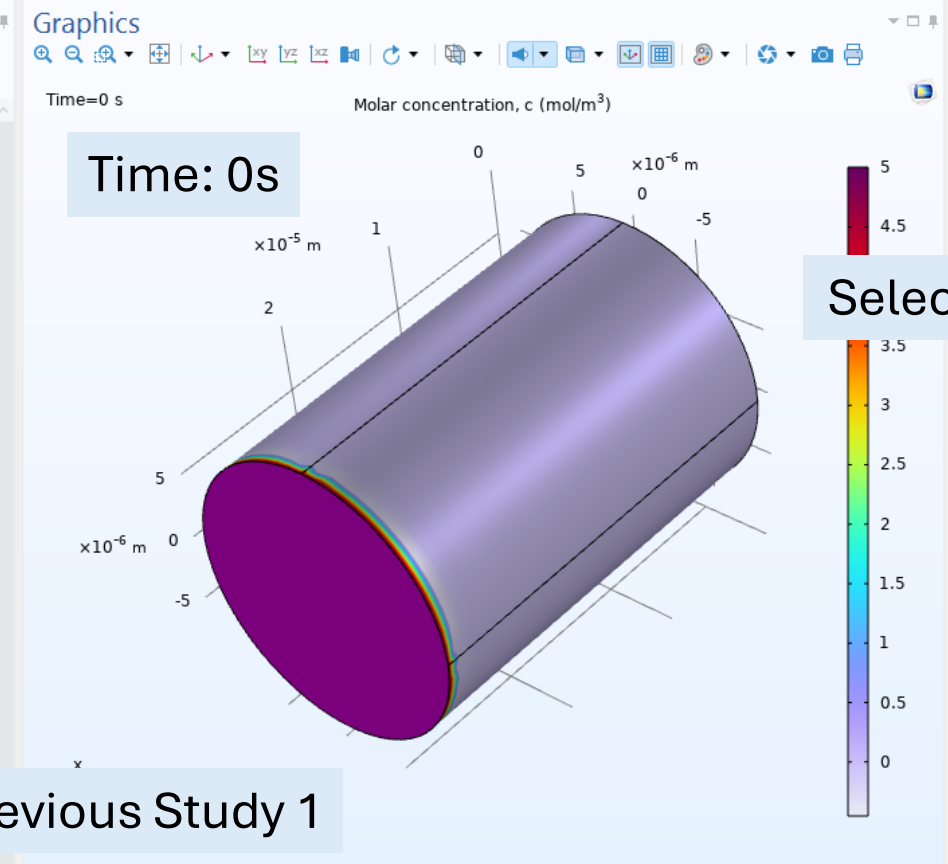
Settings: User controlled

Method: Initial expression

Study: Study 1, Stationary

Solution: Current (Solution 1 (sol1))

Selection: Automatic (single solution)



Add Study x Add Multip... x Add Material x

+ Add Study

Studies

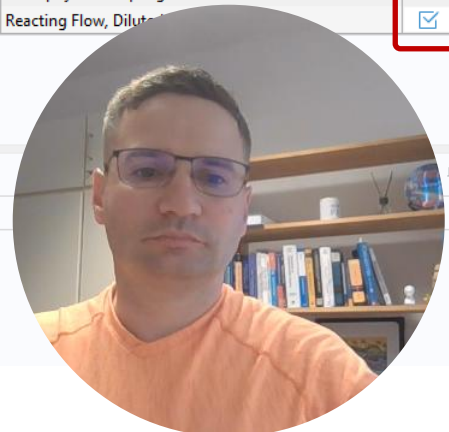
- General Studies
 - Stationary
 - Time Dependent**
 - Empty Study

Physics interfaces in study

Physics	☐	Solve
Laminar Flow (spf)	☐	
Transport of Diluted Species (tds)	☒	
Multiphysics couplings in study		
Multiphysics couplings		Solve
Reacting Flow, Diluted Species 1 (...)	☒	

Select the previous Study 1

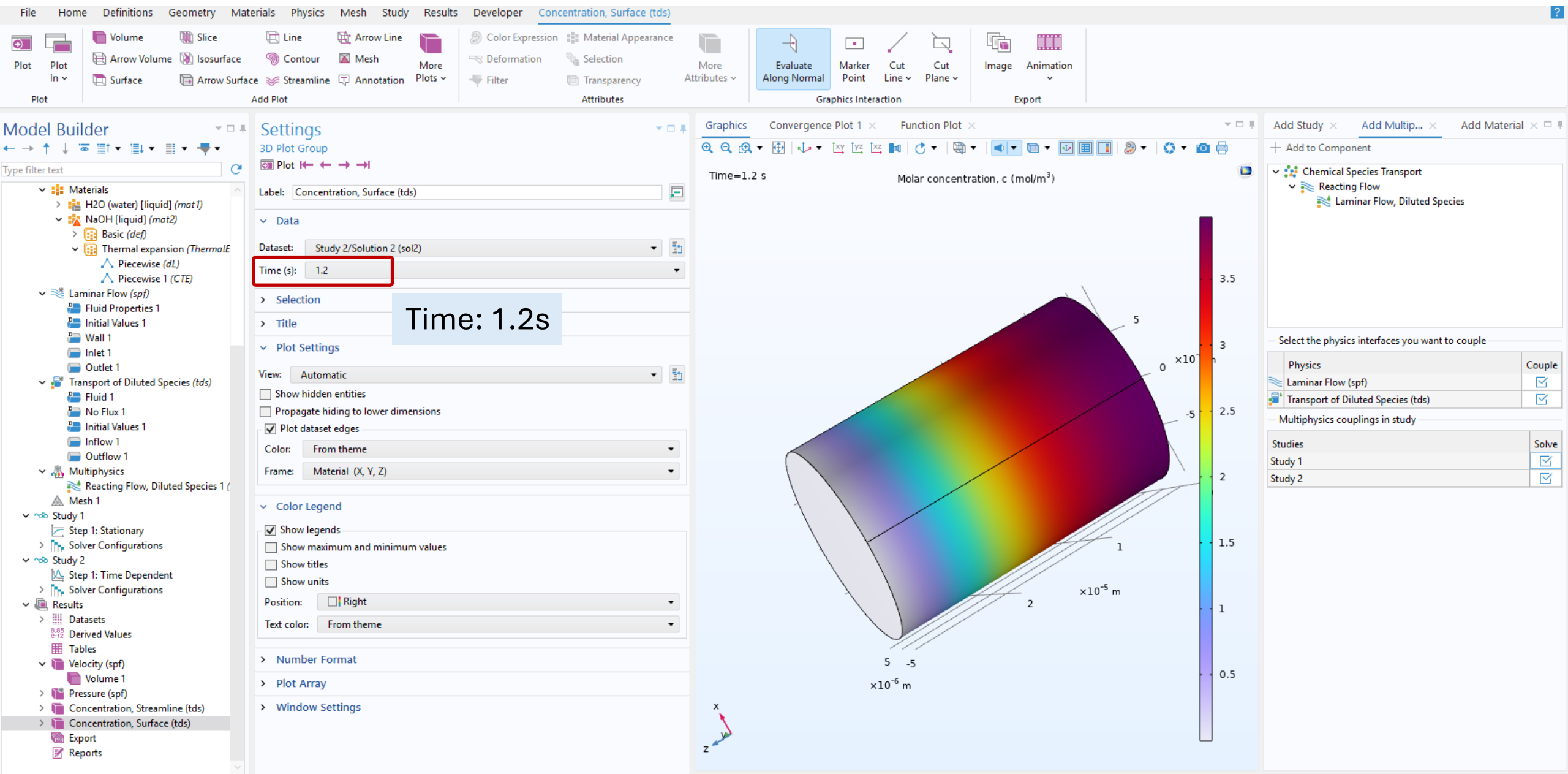
Select "Time Dependent"

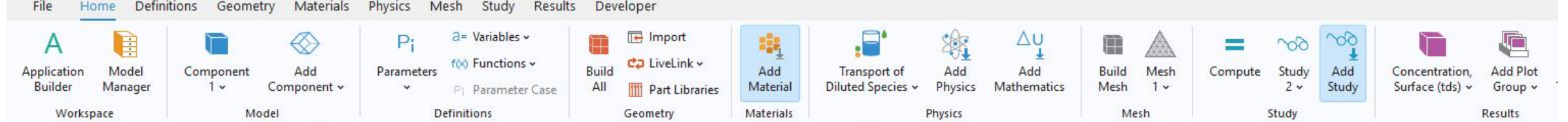


Messages x Progress Log Table 1 x

COMSOL Multiphysics 6.3.0.290

[Sep 15, 2026, 3:53 PM] Opened file: D:\papers\tutorial2026nanocom\single_pipe.mph





Model Builder

Type filter text

- Materials
 - H2O (water) [liquid] (mat1)
 - NaOH [liquid] (mat2)
 - Basic (def)
 - Thermal expansion (ThermalE)
 - Piecewise (dL)
- Land
 - Global Evaluation
 - Point Matrix Evaluation
 - Global Matrix Evaluation
 - System Matrix
- Tran
 - Average
 - Volume Average
 - Surface Average
 - Line Average
 - Integration
 - Maximum
 - Minimum
 - Measure
 - More Derived Values
- Mu
 - Evaluate All
 - Clear and Evaluate All
 - Node Group
 - Copy as Code to Clipboard
 - Settings
 - Properties and Comments
 - Help

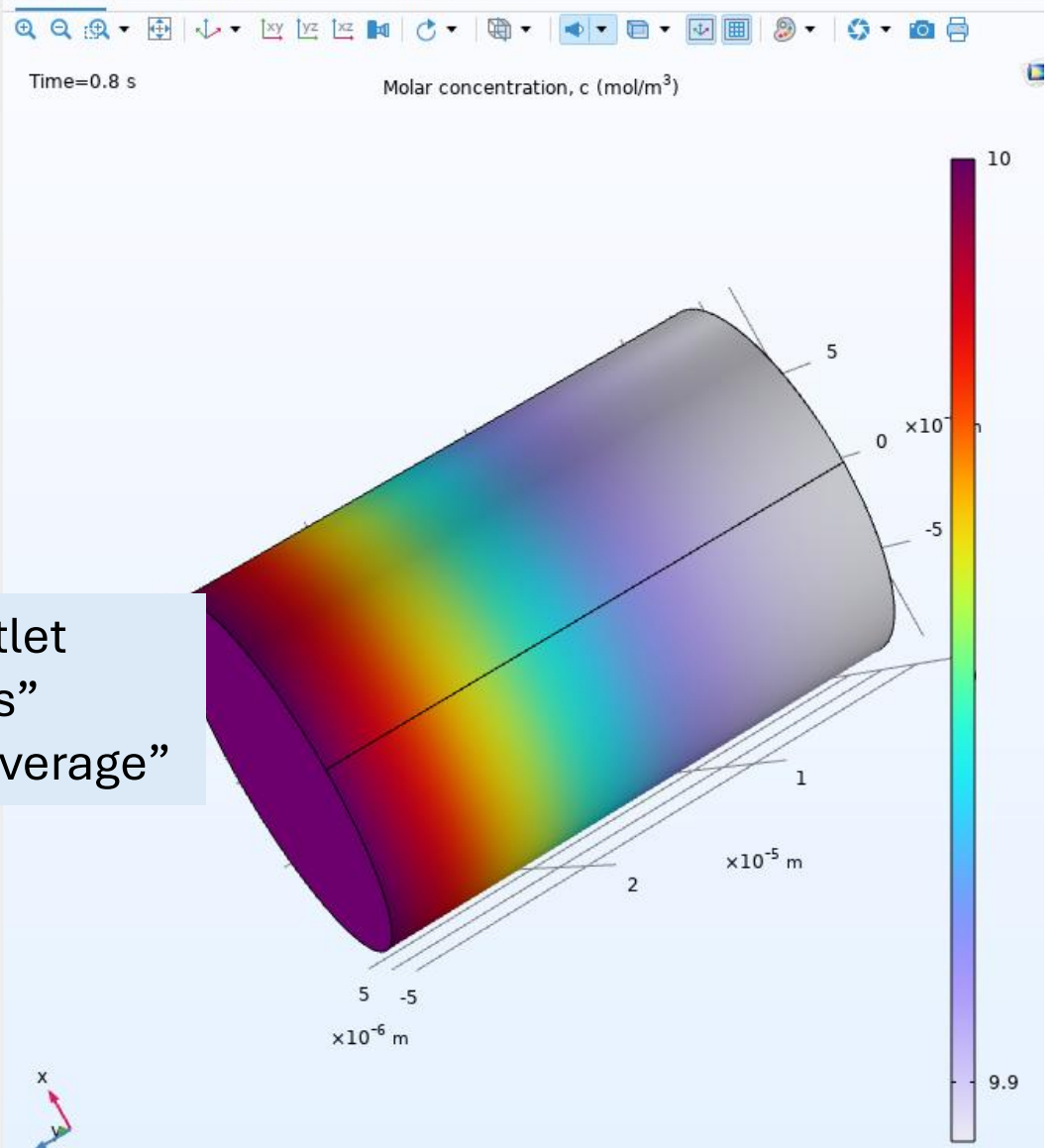
Settings

Derived Values

Average concentration at the Outlet

- Right-click on “Derived Values”
- Select “Average” → “Surface Average”

Graphics



Model Builder

Type filter text

- H2O (water) [liquid] (mat1)
- NaOH [liquid] (mat2)
 - Basic (def)
 - Thermal expansion (ThermalE)
 - Piecewise (dL)
 - Piecewise 1 (CTE)
- Laminar Flow (spf)
 - Fluid Properties 1
 - Initial Values 1
 - Wall 1
 - Inlet 1
 - Outlet 1
- Transport of Diluted Species (tds)
 - Fluid 1
 - No Flux 1
 - Initial Values 1
 - Inflow 1
 - Outflow 1
- Multiphysics
 - Reacting Flow, Diluted Species 1
- Mesh 1
- Study 1
 - Step 1: Stationary
 - Solver Configurations
- Study 2
 - Step 1: Time Dependent
 - Solver Configurations
- Results
 - Datasets
 - Derived Values
 - Surface Average 1
- Tables
- Velocity (spf)
 - Volume 1
- Pressure (spf)
- Concentration, Streamline (tds)
- Concentration, Surface (tds)
- Export
- Reports

Settings

Surface Average

Evaluate

Label: Surface Average 1

Data

Dataset: Study 2/Solution 2 (sol2)

Time selection: All

Selection

Selection: Manual



3

Expressions

Expression	Unit	Description
c	mol/m ³	Molar concentration, c

- Select Study 2 (Time-dependent)
- Select the Outflow surface (Side 3)
- Input “c” as the expression to average
- Click on Evaluate

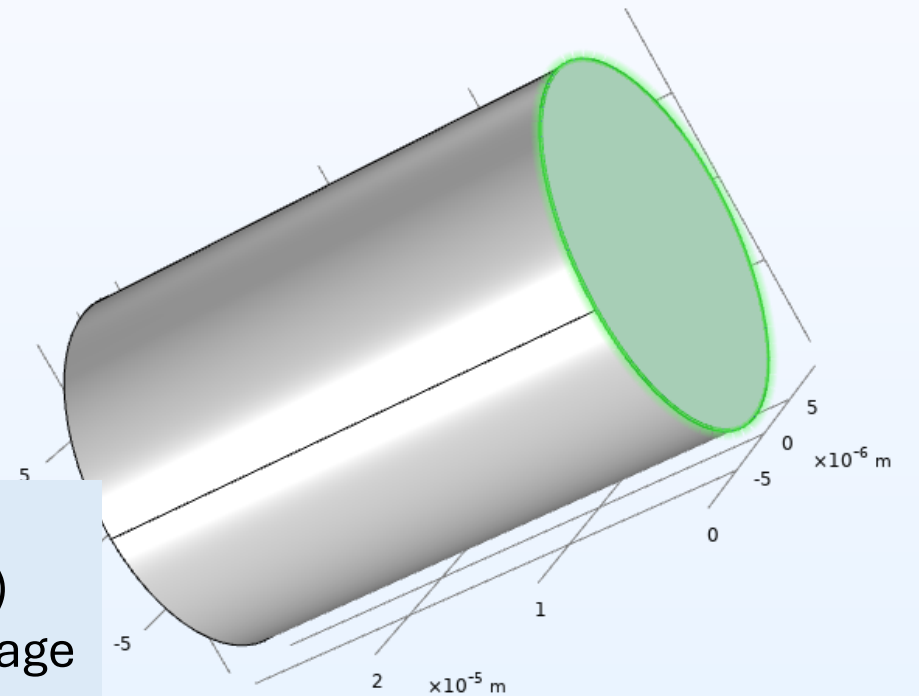
Graphics

Convergence Plot 1

Function Plot

Graphics toolbar with various icons for viewing and plotting.

→ 3 | Up



Model Builder

Type filter text

- > H2O (water) [liquid] (mat1)
- > NaOH [liquid] (mat2)
 - > Basic (def)
 - > Thermal expansion (ThermalE)
 - Piecewise (dL)
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- > Transport of Diluted Species (tds)
 - Fluid 1
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 - Inflow 1
 - Outflow 1
- > Multiphysics
 - Reacting Flow, Diluted Species 1 (
- > Mesh 1
- > Study 1
 - Step 1: Stationary
 - Solver Configurations
- > Study 2
 - Step 1: Time Dependent
 - Solver Configurations
- > Results
 - Datasets
 - > Derived Values
 - Surface Average 1
 - Tables
 - Velocity (spf)
 - Volume 1
 - Pressure (spf)
 - Concentration, Streamline (tds)
 - Concentration, Surface (tds)
 - > 1D Plot Group 5
 - Table Graph 1
 - Export

Settings

Table Graph

Plot

Label: Table Graph 1

> Data

Source: Table

Table: Table 1

☐ Row-based

x-axis data: Automatic

Plot columns: All excluding x-axis

Transformation: None

Sorting: None

☐ Plot imaginary part

> Preprocessing

> Coloring and Style

— Line style

Line: Solid

Color: Cycle

Color cycle: Default

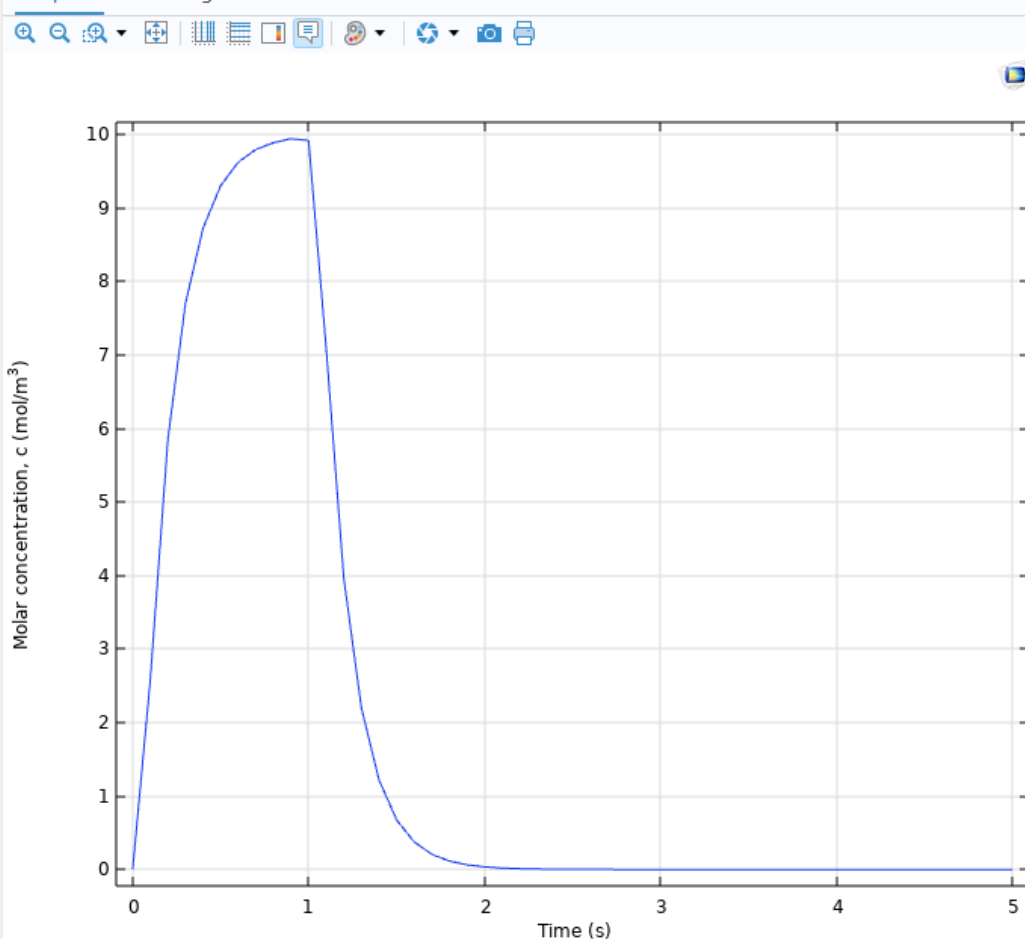
Width: Default from preferences (1)

— Line markers

Marker: None

> Legends

Graphics



Messages

Progress

Log

Table 1 ×

Time (s)	Molar concentration, c (mol/m ³)
0.0000	-1.1620E-18
0.10000	2.5788
0.20000	5.8548
0.30000	7.7052
0.40000	8.7292

Tutorial

Artificial Intelligence in Microfluidic Circuits

Speakers

Jorge Torres Gomez, Murat Kuşçu, Cansu Canbek, Volkan Can

ACM NanoCom 2026

Memorial University

Canada

September 21-23, 2026

